M580 BMENOS0300

Network Option Switch
Installation and Configuration Guide

06/2016

NHA89117.00

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All pertinent state, regional, and local safety regulations must be observed when installing and using this product. For reasons of safety and to help ensure compliance with documented system data, only the manufacturer should perform repairs to components.

When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.

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Safety Information

Important Information

NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

⚠ DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

⚠ WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

⚠ CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result** in minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

BEFORE YOU BEGIN

Do not use this product on machinery lacking effective point-of-operation guarding. Lack of effective point-of-operation guarding on a machine can result in serious injury to the operator of that machine.

⚠ WARNING

UNGUARDED EQUIPMENT

- Do not use this software and related automation equipment on equipment which does not have point-of-operation protection.
- Do not reach into machinery during operation.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

This automation equipment and related software is used to control a variety of industrial processes. The type or model of automation equipment suitable for each application will vary depending on factors such as the control function required, degree of protection required, production methods, unusual conditions, government regulations, etc. In some applications, more than one processor may be required, as when backup redundancy is needed.

Only you, the user, machine builder or system integrator can be aware of all the conditions and factors present during setup, operation, and maintenance of the machine and, therefore, can determine the automation equipment and the related safeties and interlocks which can be properly used. When selecting automation and control equipment and related software for a particular application, you should refer to the applicable local and national standards and regulations. The National Safety Council's Accident Prevention Manual (nationally recognized in the United States of America) also provides much useful information.

In some applications, such as packaging machinery, additional operator protection such as point-of-operation guarding must be provided. This is necessary if the operator's hands and other parts of the body are free to enter the pinch points or other hazardous areas and serious injury can occur. Software products alone cannot protect an operator from injury. For this reason the software cannot be substituted for or take the place of point-of-operation protection.

Ensure that appropriate safeties and mechanical/electrical interlocks related to point-of-operation protection have been installed and are operational before placing the equipment into service. All interlocks and safeties related to point-of-operation protection must be coordinated with the related automation equipment and software programming.

NOTE: Coordination of safeties and mechanical/electrical interlocks for point-of-operation protection is outside the scope of the Function Block Library, System User Guide, or other implementation referenced in this documentation.

START-UP AND TEST

Before using electrical control and automation equipment for regular operation after installation, the system should be given a start-up test by qualified personnel to verify correct operation of the equipment. It is important that arrangements for such a check be made and that enough time is allowed to perform complete and satisfactory testing.

⚠ WARNING

EQUIPMENT OPERATION HAZARD

- Verify that all installation and set up procedures have been completed.
- Before operational tests are performed, remove all blocks or other temporary holding means used for shipment from all component devices.
- Remove tools, meters, and debris from equipment.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Follow all start-up tests recommended in the equipment documentation. Store all equipment documentation for future references.

Software testing must be done in both simulated and real environments.

Verify that the completed system is free from all short circuits and temporary grounds that are not installed according to local regulations (according to the National Electrical Code in the U.S.A, for instance). If high-potential voltage testing is necessary, follow recommendations in equipment documentation to prevent accidental equipment damage.

Before energizing equipment:

- Remove tools, meters, and debris from equipment.
- Close the equipment enclosure door.
- Remove all temporary grounds from incoming power lines.
- Perform all start-up tests recommended by the manufacturer.

OPERATION AND ADJUSTMENTS

The following precautions are from the NEMA Standards Publication ICS 7.1-1995 (English version prevails):

- Regardless of the care exercised in the design and manufacture of equipment or in the selection and ratings of components, there are hazards that can be encountered if such equipment is improperly operated.
- It is sometimes possible to misadjust the equipment and thus produce unsatisfactory or unsafe operation. Always use the manufacturer's instructions as a guide for functional adjustments. Personnel who have access to these adjustments should be familiar with the equipment manufacturer's instructions and the machinery used with the electrical equipment.
- Only those operational adjustments actually required by the operator should be accessible to the operator. Access to other controls should be restricted to prevent unauthorized changes in operating characteristics.

About the Book

At a Glance

Document Scope

PlantStruxure is a Schneider Electric program designed to address the key challenges of many different types of users, including plant managers, operations managers, engineers, maintenance teams, and operators, by delivering a system that is scalable, flexible, integrated, and collaborative.

This document presents one of the PlantStruxure features, using Ethernet as the backbone around the Modicon M580 offer and connecting an M580 *local rack* and M580 *RIO drops*.

NOTE: The specific configuration settings contained in this guide are intended to be used for instructional purposes only. The settings required for your specific configuration may differ from the examples presented in this guide.

Validity Note

This document is valid for the M580 system when used with Unity Pro 11.1 or later.

The technical characteristics of the devices described in this document also appear online. To access this information online:

Step	Action
1	Go to the Schneider Electric home page *www.schneider-electric.com*.
2	In the **Search** box type the reference of a product or the name of a product range. ● Do not include blank spaces in the reference or product range. ● To get information on grouping similar modules, use asterisks (*).
3	If you entered a reference, go to the **Product Datasheets** search results and click on the reference that interests you. If you entered the name of a product range, go to the **Product Ranges** search results and click on the product range that interests you.
4	If more than one reference appears in the **Products** search results, click on the reference that interests you.
5	Depending on the size of your screen, you may need to scroll down to see the data sheet.
6	To save or print a data sheet as a .pdf file, click **Download XXX product datasheet**.

The characteristics that are presented in this manual should be the same as those characteristics that appear online. In line with our policy of constant improvement, we may revise content over time to improve clarity and accuracy. If you see a difference between the manual and online information, use the online information as your reference.

Related Documents

Title of Documentation	Reference Number
Modicon M580 Standalone System Planning Guide for Frequently Used Architectures	HRB62666 (English), HRB65318 (French), HRB65319 (German), HRB65320 (Italian), HRB65321 (Spanish), HRB65322 (Chinese)
Modicon M580 System Planning Guide for Complex Topologies	NHA58892 (English), NHA58893 (French), NHA58894 (German), NHA58895 (Italian), NHA58896 (Spanish), NHA58897 (Chinese)
Modicon M580 Hot Standby Installation and Configuration Guide	NHA58880 (English), NHA58881 (French), NHA58882 (German), NHA58883 (Italian), NHA58884 (Spanish), NHA58885 (Chinese)
Modicon M580 Hardware Reference Manual	EIO0000001578 (English), EIO0000001579 (French), EIO0000001580 (German), EIO0000001581 (Spanish), EIO0000001582 (Italian), EIO0000001583 (Chinese)
Modicon M580 Remote I/O Modules Installation and Configuration Guide	EIO0000001584 (English), EIO0000001585 (French), EIO0000001586 (German), EIO0000001587 (Spanish), EIO0000001588 (Italian), EIO0000001589 (Chinese)
Modicon M580 Change Configuration on the Fly User Guide	EIO0000001590 (English), EIO0000001591 (French), EIO0000001592 (German), EIO0000001593 (Spanish), EIO0000001594 (Italian), EIO0000001595 (Chinese)

Title of Documentation	Reference Number
Modicon M340/X80 with Unity Pro Discrete Input/Output Modules User Manual	35012474 (English), 35012475 (German), 35012476 (French), 35012477 (Spanish), 35012478 (Italian), 35012479 (Chinese)
Modicon M340/X80 with Unity Pro BMX EHC 0200 Counting Module User Manual	35013355 (English), 35013356 (German), 35013357 (French), 35013358 (Spanish), 35013359 (Italian), 35013360 (Chinese)
Grounding and Electromagnetic Compatibility of PLC Systems Basic Principles and Measures User Manual	33002439 (English), 33002440 (French), 33002441 (German), 33002442 (Spanish), 33003702 (Italian), 33003703 (Chinese)
Unity Pro, Program Languages and Structure, Reference Manual	35006144 (English), 35006145 (French), 35006146 (German), 35013361 (Italian), 35006147 (Spanish), 35013362 (Chinese)
Unity Pro System Bits and Words, Reference Manual	EIO0000002135 (English), EIO0000002136 (French), EIO0000002137 (German), EIO0000002138 (Italian), EIO0000002139 (Spanish), EIO0000002140 (Chinese)
Unity Pro Operating Modes	33003101 (English), 33003102 (French), 33003103 (German), 33003104 (Spanish), 33003696 (Italian), 33003697 (Chinese)
Unity Pro Installation Manual	35014792 (French), 35014793 (English), 35014794 (German), 35014795 (Spanish), 35014796 (Italian), 35012191 (Chinese)

Title of Documentation	Reference Number
Modicon Controllers Platform Cyber Security, Reference Manual	EIO0000001999 (English), EIO0000002001 (French), EIO0000002000 (German), EIO0000002002 (Italian), EIO0000002003 (Spanish), EIO0000002004 (Chinese)

You can download these technical publications and other technical information from our website at http://download.schneider-electric.com

Chapter 1
Introducing the M580 BMENOS0300 Network Option Switch

BMENOS0300 Module Features

Introduction

The BMENOS0300 Ethernet network option switch is a low-cost embedded switching module that simplifies the Ethernet architecture by reducing the use of external switches. You can install the module on a local or a remote BMEXBP•••• Ethernet rack in the Modicon M580 system.

NOTE: If you mount the BMENOS0300 switch on a BMX (X Bus only) backplane, the module will not power up. The module can obtain power only when installed on a BME (Ethernet) rack.

Ports

The BMENOS0300 module includes the following 100Base-T (RJ45) network ports:

- 2 network ports supporting RSTP redundant topologies
- 1 service port

The module also includes an internal backplane port, which is used to communicate to the embedded Ethernet backplane resident in the rack.

All ports support:

- 10/100 Mbps data rate
- Ethernet II frame type

The two network ports can support the following topologies:

- A DIO ring for distributed devices, including Advantys and TeSys T I/O devices
- An RIO sub-ring for (e)X80 RIO drops
- A non-redundant connection to DIO clouds consisting of distributed equipment

NOTE: Use the IP address of the CPU, and not the address of the BMENOS0300 switch, to communicate with the CPU.

The service port supports:

- Mirroring of all other ports, including the network ports, backplane port, and the internal port connecting the embedded switch to the module firmware.
- A non-looping daisy-chain of distributed equipment.
- General access, for example, by engineering tools performing firmware upgrades.

Configuration

Use the two rotary switches on the front of the BMENOS0300 module to configure the module. The position of these two switches determine the configurable module settings, other than the IP address.

- The left rotary switch configures the service port.
- The right rotary switch configures the two network ports.

The BMENOS0300 module includes a DHCP client, and receives its IP address in one of the following ways:

- From a DHCP server resident in the M580 CPU that has been configured to provide the module with its IP address.
- If no DHCP server is configured to provide an IP address to the BMENOS0300 module, the module configures itself with an IP address based on its MAC Address.

Ethernet Services

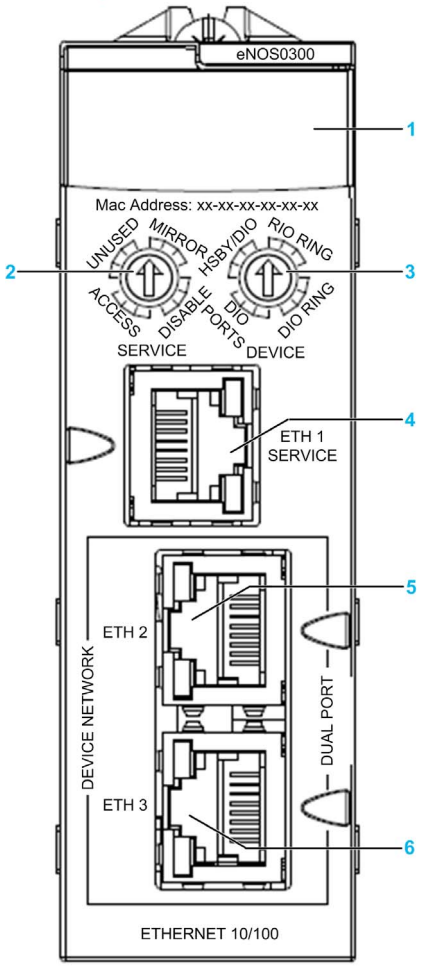
The BMENOS0300 module supports the following Ethernet services:

- Web Access HTTP Server
- Network Management SNMP agent
- Message priority QoS tagging
- Diagnostic embedded web pages
- DHCP client
- FTP

Chapter 2
BMENOS0300 Physical Description

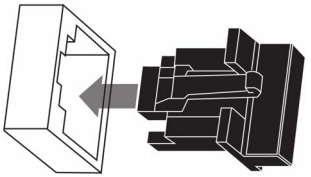
Module Description

Physical Description



1	LED array *(see page 62)*: ● RUN ● ERR ● MS ● NS
2	Rotary switch *(see page 24)* for configuring the ETH 1 service port: ● MIRROR ● DISABLE ● ACCESS ● UNUSED
3	Rotary switch *(see page 24)* for configuring the two device network ports (ETH 2 and ETH 3): ● RIO RING ● DIO RING ● DIO PORTS ● HSBY/DIO
4	ETH 1: Service port (Ethernet)
5	ETH 2: Device network port (Ethernet)
6	ETH 3: Device network port (Ethernet)

To help keep dust out of unused Ethernet ports, cover the ports with a stopper:



Chapter 3
BMENOS0300 Specifications

Module Specifications

Standards and Certifications

For this information, refer to the Standards and Certifications *(see Modicon M580, Hardware, Reference Manual)* topic in the *Modicon M580 Hardware Reference Manual*.

Operational and Environmental Recommendations

For this information, refer to the Standards and Certifications *(see Modicon M580, Hardware, Reference Manual)* topic in the *Modicon M580 Hardware Reference Manual*.

Current and Power Profiles

The BMENOC03•1 module offers the following current and power profile:

Parameter	Specification
Voltage rail	24 V_BAC
Current at 25° C (77° F) ambient	135 mA
Power at 25° C (77° F) ambient	3.3 W
Current at 60° C (140° F) ambient	185 mA
Power at 60° C (140° F) ambient	4.5 W

Chapter 4
Installing the BMENOS0300 Module

Mounting a BMENOS0300 Module on an M580 Rack

Introduction

The BMENOS0300 Ethernet network option switch can be installed on a either a local or remote Ethernet backplane in the M580 system.

NOTE:

- Mount the BMENOS0300 switch only on a BME (Ethernet) backplane. The module powers up on a BME (Ethernet) backplane, but does not power up on a BMX (X-Bus) backplane.
- Schneider Electric recommends that you limit your system to two BMENOS0300 modules in an M580 rack for optimal system performance. If you install more than two BMENOS0300 modules per rack, calculate the bandwidth of your system so you do not overload the 100 Mbit/second backplane capacity.

Before Installing a Module

Before installing the module, remove the protective cap from the module connector on the rack.

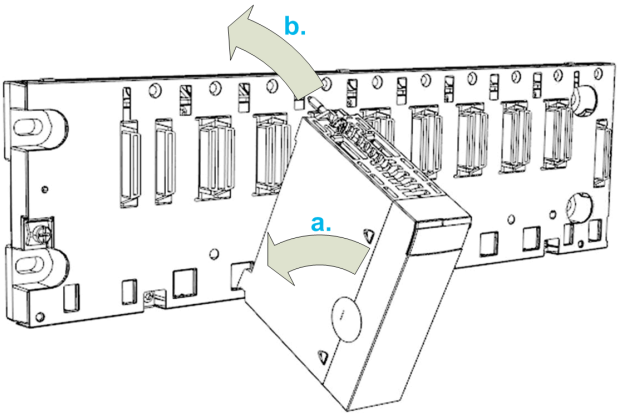
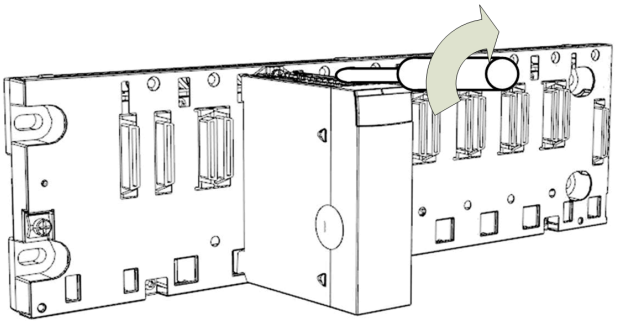
Selecting a Backplane

Install the BMENOS0300 module in a single slot on one of these Ethernet backplanes:

Backplane	Description
BMEXBP0400[1]	4-slot Ethernet backplane
BMEXBP0400(H)[1]	4-slot hardened Ethernet backplane
BMEXBP0800[1]	8-slot Ethernet backplane
BMEXBP0800(H)[1]	8-slot hardened Ethernet backplane
BMEXBP1200[1, 2]	12-slot Ethernet backplane
BMEXBP1200(H)[1, 2]	12-slot hardened Ethernet backplane
BMEXBP0602 (H)	10-slot hardened Ethernet and X Bus backplane
BMEXBP1002 (H)[3]	6-slot hardened Ethernet and X Bus backplane
1. In a local rack, slots 0 and 1 are reserved for the CPU. 2. In the 12-slot Ethernet backplane, slots 2, 8, 10, and 11 are X Bus only slots. 3. In the 10-slot Ethernet backplane, slots 2 and 8 are X Bus only slots.	

Installing the Module on the Rack

Mount the module in a single slot on the backplane:

Step	Action
1	Turn off the power supply to the rack.
2	Remove the protective cover from the module interface on the rack.
3	*a*: Insert the locating pins on the bottom of the module into the corresponding slots in the rack.  *b*: Use the locating pins as a hinge and pivot the module until it is flush with the rack. (The connector on the back of the module inserts into the Ethernet connector on the rack.)
4	Tighten the retaining screw to hold the module in place on the rack:  Tightening torque: 1.5 N•m max. (1.11 lb-ft)

Grounding Considerations

⚠ DANGER

ELECTRICAL SHOCK HAZARD

- Switch off the power supply to the rack at both ends of the connection before inserting or removing an Ethernet cable.
- Use suitable insulation equipment when inserting or removing all or part of this equipment.

Failure to follow these instructions will result in death or serious injury.

Do not apply power to the BMENOS0300 module until connections are made at both ends of the Ethernet cable. For example, connect the cable to both the module and another device (for example, an adapter module) before you turn on the power.

Use fiber optic cable and BMXNRP020• fiber converter modules to establish a communications link when it is not possible to equalize the potential between the two grounds.

NOTE: Refer to the ground protection information provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual* and *Control Panel Technical Guide, How to protect a machine from malfunctions due to electromagnetic disturbance*.

Replacing a Module

Any BMENOS0300 module on the rack can be replaced at any time with another module with compatible firmware. The replacement module obtains its configuration settings as follows:

- Operating parameters (except for IP addressing settings) are set by the positions of the two rotary switches on the front of the module. When replacing a module, set the rotary switches on the replacement module to the same positions as the rotary switches on the original module.
- IP address settings for the BMENOS0300 module are configured in the CPU, in its role as DHCP server. Use Unity Pro to configure the **Services → Address Server** page of the CPU DTM and add each BMENOS0300 module to the list of DHCP clients served IP address settings by the CPU. Refer to the topic Determining the Device Name *(see page 28)*.

Hot Swap

From the system point of view, during a hot swap of the BMENOS0300 module, when the module is removed the I/O values go to fallback values. When the new module is inserted and powers up, the I/O values reset to their values before the hot swap. Refer to the topic BMENOS0300 DDDT *(see page 73)* for information about module DDDT values.

Chapter 5
Configuring the BMENOS0300 Module

Overview

This chapter describes the configuration of the module, including its external Ethernet ports and IP address.

The BMENOS0300 module is compatible with the following related product versions

Product	Version
Unity Pro	Version 11.0 or higher
M580 CPU firmware	Version 2.10 or higher
BMEXBP*xxxx* backplane firmware	Version 1.0 or higher
ConneXium Network Manager software	Version 6.0 or higher

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
Configuring BMENOS0300 Ethernet Ports	24
Assigning an IP Address to the BMENOS0300 Module	26

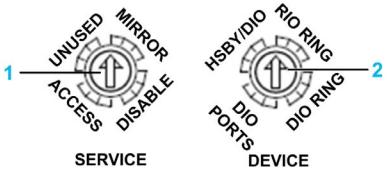
Configuring BMENOS0300 Ethernet Ports

Configuration Via Rotary Switches

Configuration of the BMENOS0300 module port settings is performed exclusively by means of the two rotary switches on the front of the module:

- The left switch configures the service port (ETH 1).
- The right switch configures the device network ports (ETH 2 and ETH 3).

Use a small flat head screwdriver to turn each rotary switch.



1 ETH 1 Service Port configuration switch
2 ETH 2 & 3 Device Network ports configuration switch

Use the rotary switches to configure the Ethernet ports, as described below.

Service Rotary Switch Configuration Positions

Use the Service rotary switch to configure the service port (ETH 1), as follows:

Switch Position	Configuration Setting Supports...
MIRROR	Port mirroring. Sends to ETH 1 a copy of Ethernet packets sent over all other Ethernet ports, including: ● ETH 2 ● ETH 3 ● Backplane port ● Internal port (connecting the three Ethernet ports on the front of the module to the module firmware)
ACCESS	General Ethernet communication to Ethernet devices, including: ● A PC running Unity Loader, for the purpose of performing a firmware upgrade. ● Distributed equipment in a non-looping Ethernet daisy chain.
DISABLE	No Ethernet packets pass through this port. This is the default setting.
UNUSED	The same behavior as the ACCESS switch position.

Device Rotary Switch Positions

Use the Device rotary switch to configure the network ports (ETH 2 and ETH 3).

NOTE: To enable RSTP, declare the BMENOS0300 module in Unity Pro. To declare the module in Unity Pro, add it to a rack in the **PLC Bus** or **EIO Bus** window.

Switch Position	Configuration Setting Supports...
HSBY/DIO	A non-deterministic distributed equipment Ethernet ring and RSTP, which employs two BMENOS0300 modules in a Hot Standby system as follows: ● One module located in the primary Hot Standby local rack ● One module located in the standby Hot Standby local rack. When HSBY/ DIO is selected: ● The BMENOS0300 module in the Hot Standby rack with CPU “A” becomes the RSTP root bridge on the DIO ring, with a bridge identifier priority value set to 0. ● The BMENOS0300 module in the Hot Standby rack with CPU “B” has its bridge identifier priority value set to 4096 (decimal). If you select HSBY/DIO, set the bridge identifier priority settings of the other devices in the DIO ring to a value greater than 0 so they will not attempt to become the root bridge.
RIO RING	A deterministic Ethernet RIO ring and RSTP. When RIO Ring is selected, the BMENOS0300 module becomes the RSTP root bridge on the RIO ring, with a bridge identifier priority value set to 0. If you select RIO Ring, set the bridge identifier priority settings of the other devices in the RIO ring to a value greater than 0 so they will not attempt to become the root bridge.
DIO RING	A non-deterministic distributed equipment Ethernet ring and RSTP. When DIO Ring is selected: ● The BMENOS0300 module bridge identifier priority value is fixed at 61440 (decimal). ● The port cost of each network port (ETH 2 and ETH 3) are set to 200,000,000. If you select DIO Ring, assign a bridge identifier priority to another device on the ring to a value less than 61440. As a result: ● A device on the ring other than the BMENOS0300 module is the root bridge. ● One of the network ports of the BMENOS0300 module is set to discarding state. This makes it possible to determine the health of the DIO ring by checking the status (discarding or forwarding) of module network ports.
DIO PORTS	Distributed equipment connected as DIO cloud or non-looping daisy chain. Does not support RSTP. This the default setting.

Applying Configuration Settings

Configuration settings take effect when the BMENOS0300 module powers up. Typically, you will configure switch settings when power to the module is turned off. Then, when you next apply power to the module, the settings you made are applied.

NOTE: Changing the switch positions while the BMENOS0300 module is operating has no immediate effect on the module. Any changes to switch positions made while the module is operating take affect only after power is turned off, and then turned back on.

Assigning an IP Address to the BMENOS0300 Module

Two Methods of Assignment

The BMENOS0300 module is a DHCP client that receives its IP address in one of two ways:

- From a DHCP server resident in the M580 CPU.
- If no DHCP server is configured to provide an IP address to the BMENOS0300 module, the module configures itself with an IP address based on its MAC Address.

BMENOS0300 as DHCP Client

To configure the BMENOS0300 module to receive IP address settings from – and be scanned by – the CPU, follow these steps:

Step	Action	
1	With your project offline in Unity Pro, add a BMENOS0300 module to the rack:	
	a	In the **Project Browser** (structural view), select: **Configuration → PLC Bus**.
	b	Right click and select **Open**. The **PLC bus** window opens.
	c	Select a slot in rack where you want to add a BMENOS0300 module, right click, then select **New Device...**. The **New Device** window opens.
	d	In the **New Device** window, select **Modicon M580 local drop → Communication → BME NOS 0300**, then select **OK**.
	e	Click **Save**.
2	Open the **DTM Browser** to view the device DTM you are about to add:	
	a	In **DTM Browser**, right click the CPU and select **Open**.
	b	In the CPU DTM, select **Services → Address Server**. **NOTE:** You can add the new module as a DHCP client in either of two ways: ● In the **Manually Added Devices** area of the **Address Server** window, click **Add**, then enter IP addressing parameters for the BMENOS0300 module. Subsequently, the DHCP server in the CPU will send the module the IP address settings you input. However, the CPU will not scan the module. ● Follow steps 3...6 set forth below, to add the BMENOS0300 module to the list of **Automatically Added Devices**. Subsequently, the DHCP server in the CPU will send the module the IP address settings you input, and the CPU will scan the module parameters described in its DDDT *(see page 73)*.

Step	Action	
3	Add a BMENOS0300 module to be scanned by and to receive right IP address settings from the CPU:	
	a	in the **DTM Browser**, right click the CPU and select **Add...**.
	b	Select **Advanced Generic EDS**, then click **Add DTM**.
	c	In the **Properties of device** dialog, type in a name and click **OK**. **NOTE:** The name entered in this step is a variable you can assign to the module that has relevance to your application (for example, “Local NOS” or “NOS_1‘”. It is not the **Device Name** identifier value used by the DHCP server to identify the device. The new; device is added to list of **Automatically Added Devices** in the **Services → Address Server** window of the CPU.
4	Assign IP address settings to the BMENOS0300 module:	
	a	In CPU DTM, select the new module in the **Device List**, then select the **Address Setting** tab.
	b	In the **IP Configuration** area, accept the default settings, or enter custom IP address settings. **NOTE:** f you are editing settings, confirm that the BMENOS0300 module is assigned to the same subnet as the CPU.
	c	In the **Address Server** section, enter the following settings: ● **DHCP for this device**: Select **Enabled**. ● **Identified by**: Select **Device Name**. ● **Identifier**: Enter the device name *(see page 28)* value.
	d	Click **Apply** to save your edits.
5	Configure the connection for the BMENOS0300 module to the CPU:	
	a	In **DTM Browser** navigate to the BMENOS0300 DTM, right click, then select **Open**.
	b	In the left pane of the BMENOS0300 DTM, select the connection. By default, the connection is of the type **Exclusive Owner**.
	c	Select the **General** tab, and edit the following settings: ● **RPI**: Accept the default (30 ms) or enter a different value. ● **Input size**: Enter **16** (bytes). ● **Input mode**: Select **Point to Point**.
	d	Select the **Configuration Settings** tab, and edit the following settings: ● **Input instance**: Enter **101**. ● **Output instance**: Enter **198**. ● **Configuration instance**: Enter **100**. ● **Configuration**: Enter **00 00 00 00 00 00 00 00**.
	e	Click **Apply** to save your edits.
6	Click **Save**.	

Determining the Device Name

The device name identifier for the BMENOS0300 module is a concatenation of the following information that can be derived from the module's position in a rack:

```
head module abbreviation_slot number_module type
```

These three criteria can have the following values:

Criteria	Possible values
Head module abbreviation	For local racks: ● Mx80: Standalone M580 CPU ● M58A: Hot Standby CPU designated as A ● M58B: Hot Standby CPU designated as B For remote racks: ● Cxxx: CRA at drop xxx (for example "C001")
Slot number	Two digit slot number position of the BMENOS0300 module in the local or remote rack (for example "04")
Module type	A constant string value set to: ● BMENOS

Examples of device names include:

- Mx80_04_BMENOS: located in the fifth slot (04) of a local rack with a standalone M580 CPU.
- M58B_03_BMENOS: located in the fourth slot (03) of a local rack with a Hot Standby M580 CPU designated as CPU "B".
- C002_06_BMENOS: located in the seventh slot (06) of a remote rack identified as drop 2 with a CRA as its head module.

Default IP Address Assignment

If the DHCP server in the CPU is not configured to supply the BMENOS0300 module with an IP address, the module will assign itself an IP address based on its MAC address, as follows:

10.10.*x.x*

Where *x.x* represents the decimal version of the last two (fifth and sixth) octets of the MAC Address.

NOTE: After the module assigns itself an IP address based on its MAC address, it continues to broadcast requests for an IP address as a DHCP client.

Chapter 6
Supported Topologies

Overview

This chapter presents several topologies that support the use of the BMENOS0300 network option switch.

NOTE: The topologies presented in this chapter are recommended and supported by Schneider Electric. This chapter does not describe every possible topology, and specifically excludes topologies that could create broadcast storms.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
6.1	Standalone PAC, Local Rack BMENOS0300 Topologies	30
6.2	Standalone PAC, RIO Drop Topologies for the BMENOS0300	36
6.3	Hot Standby PAC, Local Rack BMENOS0300 Topologies	45
6.4	Hot Standby PAC, RIO Drop Topologies for the BMENOS0300	47

Section 6.1
Standalone PAC, Local Rack BMENOS0300 Topologies

What Is in This Section?

This section contains the following topics:

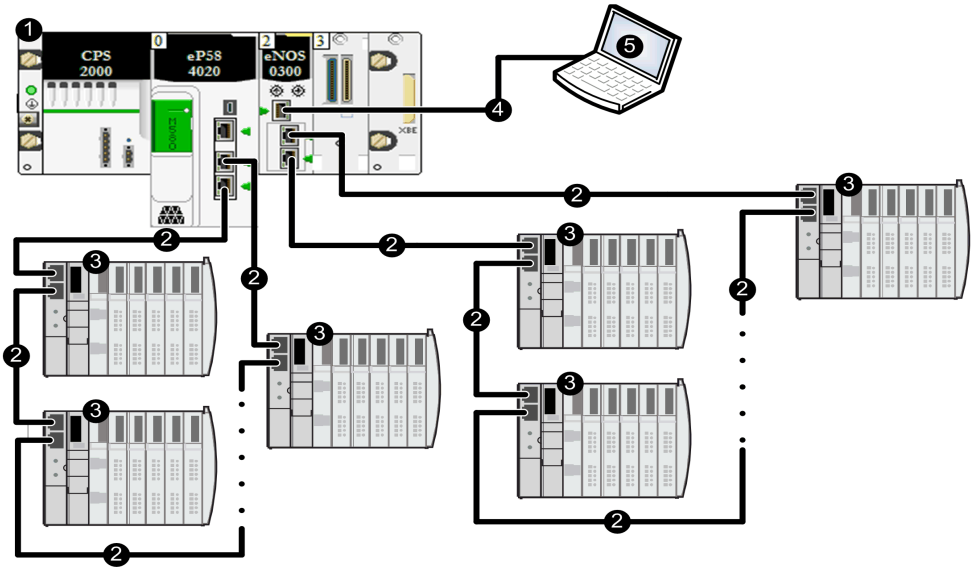
Topic	Page
BMENOS0300 in Standalone PAC Local Rack in support of DIO Ring	31
BMENOS0300 in Standalone PAC Local Rack with Two BMENOC03•1 Modules Supporting RIO and DIO Rings	32
Two BMENOS0300 Modules in Standalone PAC Local Rack with BMENOC03•1 Module in support of dual DIO Rings	34

BMENOS0300 in Standalone PAC Local Rack in support of DIO Ring

Simple DIO Ring

In this topology, the BMENOS0300 network option switch is added to the main rack. This makes it possible to add a second DIO ring (in addition to the DIO ring connected to the CPU), and thereby extend the number of distributed equipment devices beyond the 40 devices supported by a single DIO ring. The ports of the BMENOS0300 module are used as follows:

- The two network ports are configured for a **DIO Ring**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet DIO ring
3 Distributed equipment
4 Ethernet non-looping daisy chain
5 Engineering tool resident on PC

NOTE:

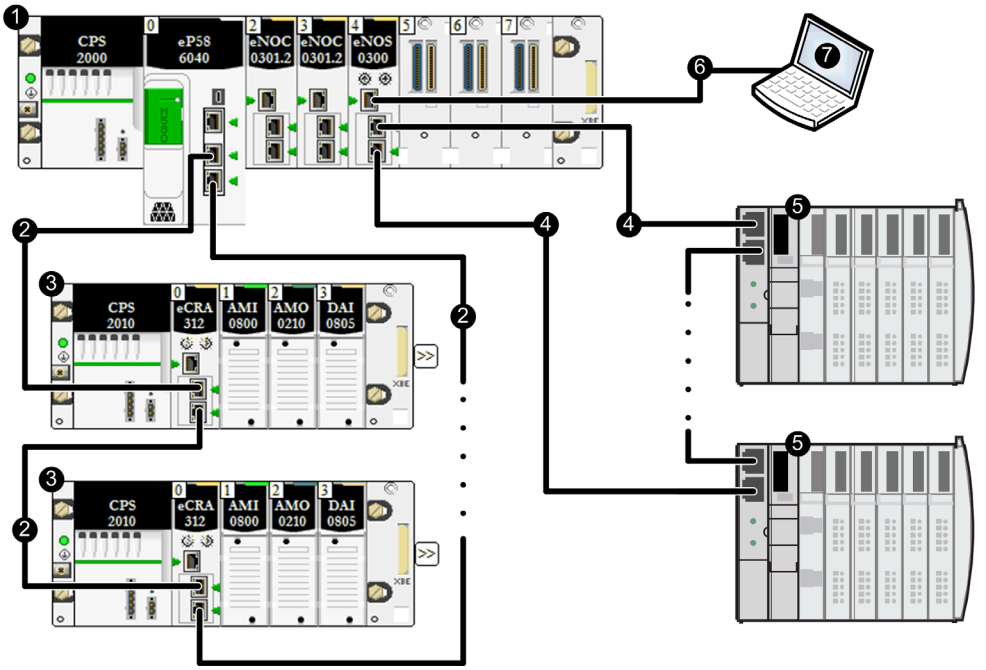
- To achieve acceptable recovery times, limit each DIO design to a single DIO main ring without sub-rings.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.

BMENOS0300 in Standalone PAC Local Rack with Two BMENOC03•1 Modules Supporting RIO and DIO Rings

RIO Main Ring and DIO Ring

In this topology, the BMENOS0300 network option switch is used in place of a BMENOC03•1 to reduce cost. The ports of the BMEP58•040 CPU and the BMENOS0300 module are used as follows:

- The two network ports on the CPU support an RIO main ring.
- The two network ports on the BMENOS0300 switch are configured for a **DIO Ring**.
- The service port on the BMENOS0300 module is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet main RIO ring
3 (e)X80 RIO drop
4 Ethernet DIO ring
5 Distributed equipment
6 Ethernet non-looping daisy chain
7 Engineering tool resident on PC

NOTE:

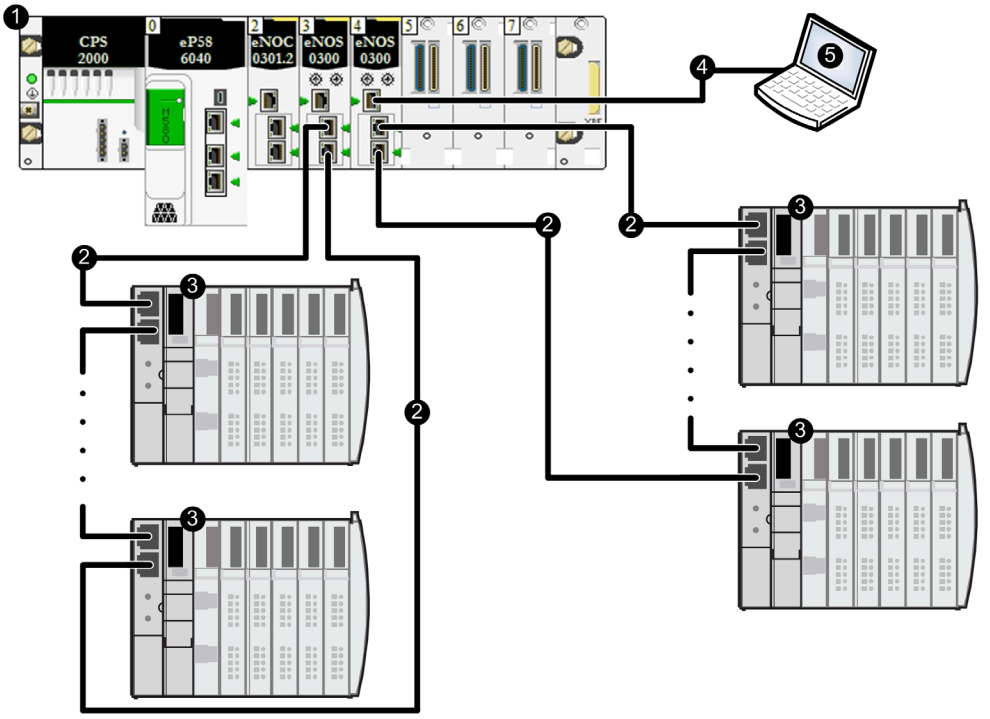
- To achieve acceptable recovery times, limit each DIO design to a single DIO main ring without sub-rings.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.

Two BMENOS0300 Modules in Standalone PAC Local Rack with BMENOC03•1 Module in support of dual DIO Rings

Dual DIO Rings

In this topology, two BMENOS0300 network option switches are used in place of BMENOC03•1 modules to reduce cost. The ports of two BMENOS0300 network option switches are configured as follows:

- The two network ports of each BMENOS0300 switch are configured for a **DIO Ring**.
- The service port of one BMENOS0300 switch is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet DIO ring
3 Distributed equipment
4 Ethernet non-looping daisy chain
5 Engineering tool resident on PC

NOTE:

- To achieve acceptable recovery times, limit each DIO design to a single DIO main ring without sub-rings.
- RSTP limits the size of each DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.
- Depending on the CPU you are using, the CPU can support up to a maximum number of 128 switched devices on all DIO rings.

Section 6.2
Standalone PAC, RIO Drop Topologies for the BMENOS0300

What Is in This Section?

This section contains the following topics:

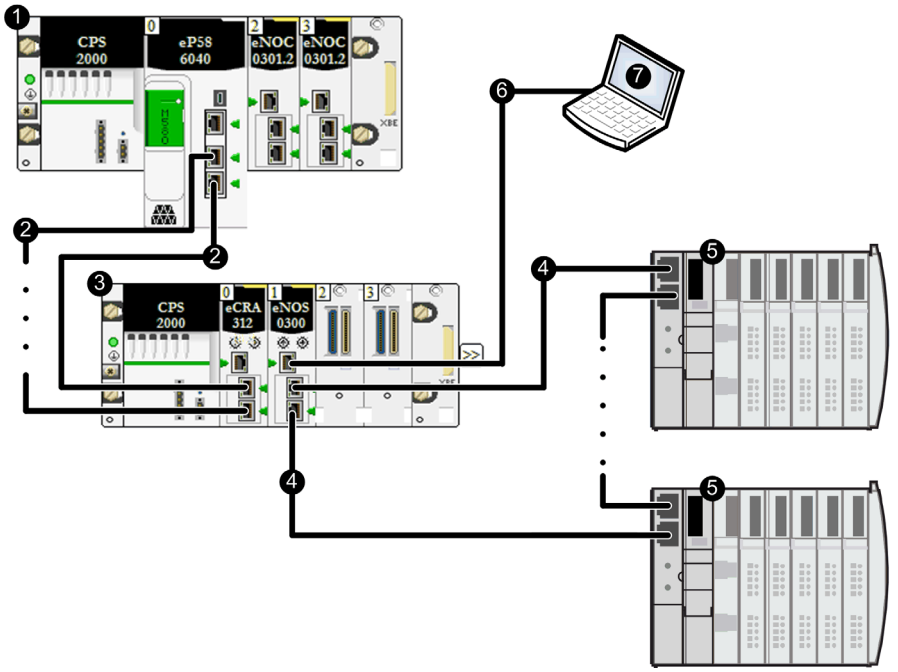
Topic	Page
BMENOS0300 in RIO Drop of Standalone PAC in support of DIO Ring	37
BMENOS0300 in RIO Drop of Standalone PAC in support of DIO Clouds	39
BMENOS0300 in RIO Drop of Standalone PAC in support of RIO Sub-Ring	41
Two BMENOS0300 Modules in RIO Drop of Standalone PAC Supporting RIO Sub-Ring and DIO Ring	43

BMENOS0300 in RIO Drop of Standalone PAC in support of DIO Ring

Simple DIO Ring

In this topology, the BMENOS0300 network option switch is used in place of a BMENOC03•1 to reduce cost. The ports of the BMENOS0300 module are used as follows:

- The two network ports are configured for a **DIO Ring**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet RIO main ring
3 (e)X80 RIO drop
4 Ethernet DIO ring
5 Distributed equipment
6 Engineering tool resident on PC

NOTE:

- To achieve acceptable recovery times, limit each DIO design to a single DIO main ring without sub-rings.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.

BMENOS0300 in RIO Drop of Standalone PAC in support of DIO Clouds

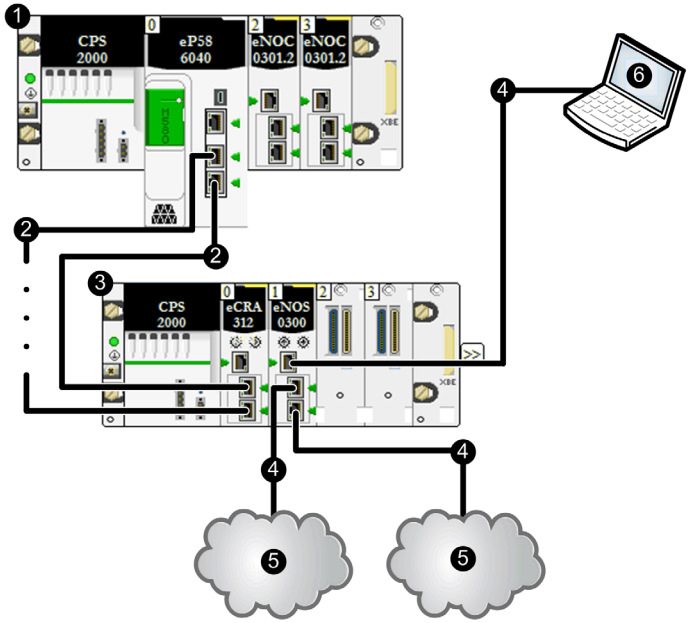
DIO Clouds

In this topology, the BMENOS0300 network option switch is used in place of a BMENOC03•1 to reduce cost. The ports of the BMENOS0300 network option switch are used as follows:

- The two network ports are configured as **DIO Ports**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.

⚠ CAUTION
RISK OF BROADCAST STORM Do not connect an Ethernet device in one DIO cloud to an Ethernet device in another DIO cloud. Connecting these devices can result in the occurrence of a broadcast storm, which can overload the network and delay or prevent intended network communications. **Failure to follow these instructions can result in injury or equipment damage.**

NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet RIO main ring
3 (e)X80 RIO drop
4 Ethernet daisy chain

5 DIO cloud
6 Engineering tool resident on PC

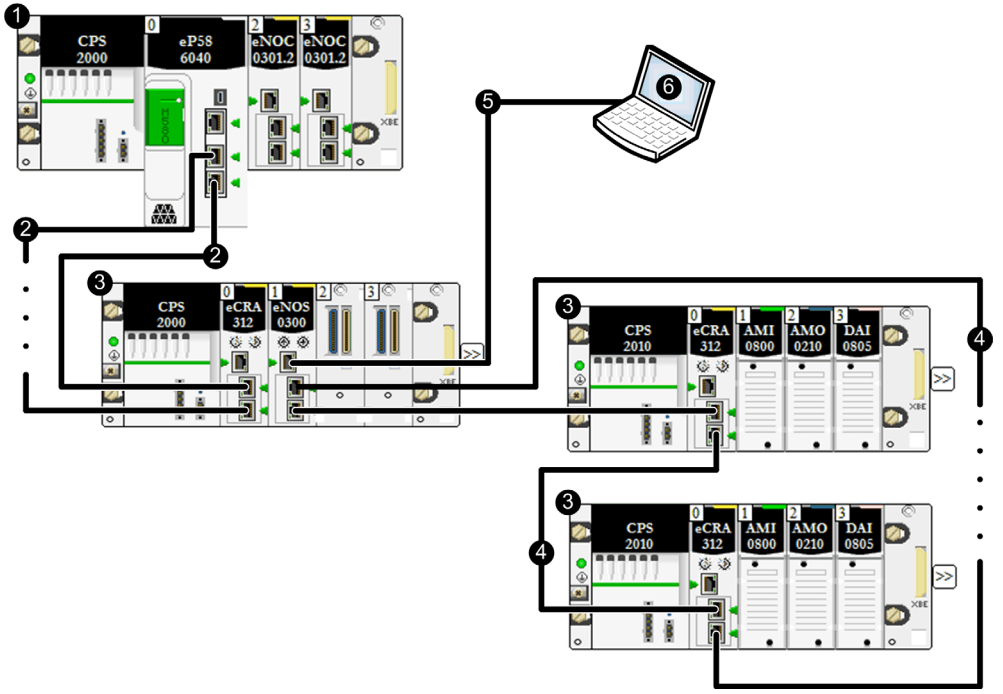
NOTE: Because RSTP is not enabled, there is no RSTP-based maximum device limit.

BMENOS0300 in RIO Drop of Standalone PAC in support of RIO Sub-Ring

Simple RIO Ring

In this topology, the BMENOS0300 network option switch is used in place of an external dual ring switch (DRS) to reduce cost. The BMENOS0300 network option switch resides in an RIO drop. Its ports are used as follows:

- The two network ports are configured for a **RIO Ring**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Standalone PAC on local rack
2 Ethernet RIO main ring
3 (e)X80 RIO drop
4 Ethernet RIO sub-ring
5 Ethernet daisy chain
6 Engineering tool resident on PC

NOTE:
To achieve desired RSTP-based network response time targets, the longest path that an Ethernet RIO packet would have to travel in the case of a cable break is 32 switched devices, including:

- the CPU
- the remote adapter (BMECRA312•0)
- the embedded switch in the Ethernet rack on which the BMENOS0300 is installed
- the BMENOS0300

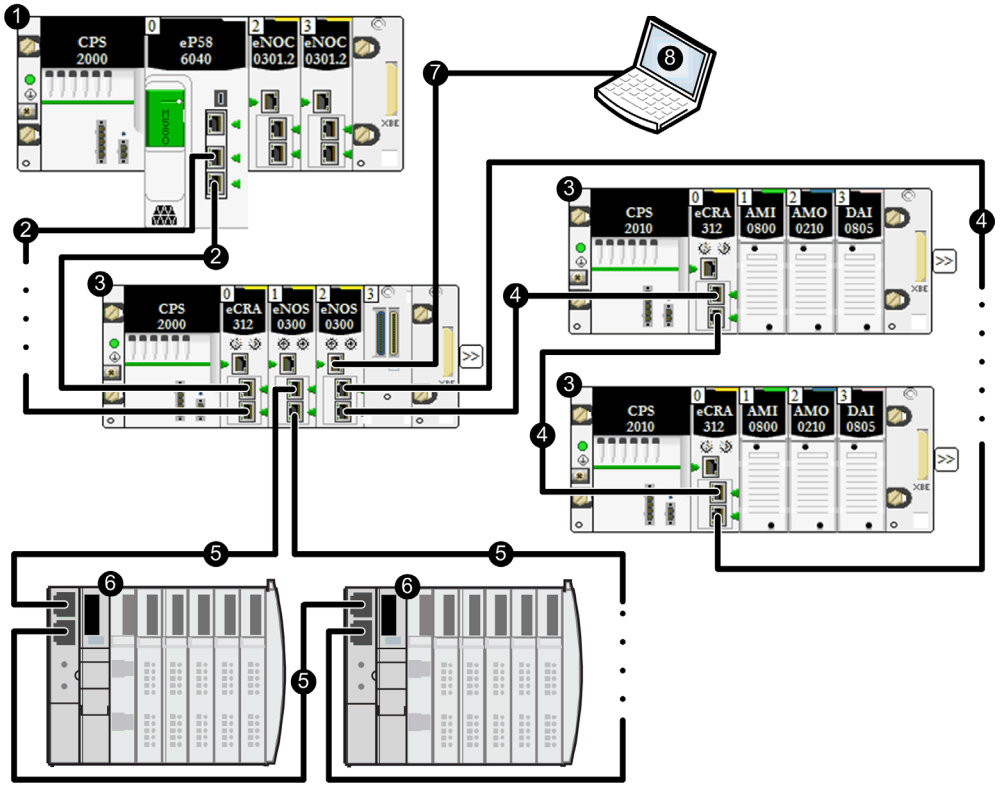
NOTE: When you set the BMENOS0300 network rotary switch to **RIO Ring**, configure RSTP bridge identifier priority settings for other devices in the RIO sub-ring to values greater than 0. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **RIO Ring**.

Two BMENOS0300 Modules in RIO Drop of Standalone PAC Supporting RIO Sub-Ring and DIO Ring

RIO Sub-Ring and DIO Ring

In this topology, two BMENOS0300 network option switches are used in place of external dual ring switches (DRSs) to reduce cost. The two BMENOS0300 network option switch modules reside in an RIO drop. The ports of these modules are used as follows:

- Two network ports of one BMENOS0300 module are configured for an **RIO Ring**, and support an RIO sub-ring.
- The service port of the same module is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.
- The two network ports of the other BMENOS0300 module are configured for a **DIO Ring**.



1 Standalone PAC on local rack
2 Ethernet main RIO ring
3 (e)X80 RIO drop
4 Ethernet RIO sub-ring

5 Ethernet DIO ring
6 Distributed equipment
7 Ethernet daisy chain
8 Engineering tool resident on PC

NOTE:

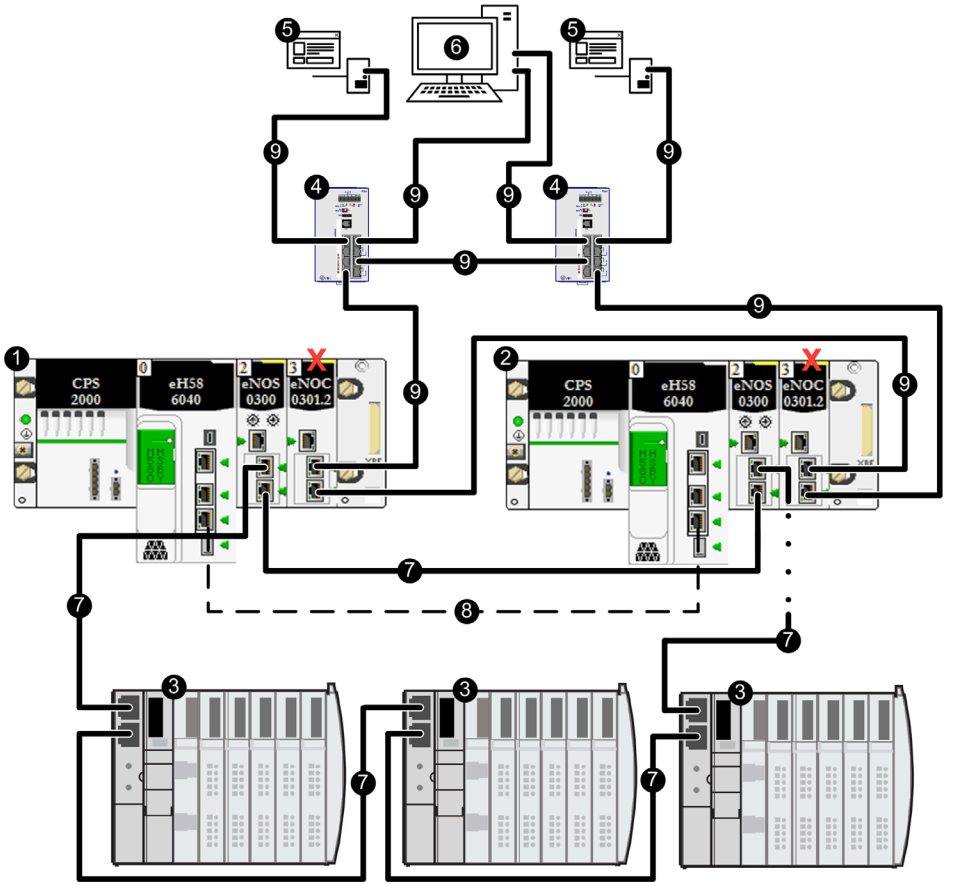
- To achieve desired RSTP-based network response time targets, the longest path that an Ethernet RIO packet would have to travel in the case of a cable break is 32 switched devices, including:
  - the CPU
  - the remote adapter (BMECRA312•0)
  - the embedded switch in the Ethernet rack on which the BMENOS0300 modules are installed
  - the BMENOS0300 module
- When you set the BMENOS0300 network rotary switch to **RIO Ring**, configure the RSTP bridge identifier priority settings of other devices in the RIO sub-ring to values greater than 0. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **RIO Ring**.
- To achieve acceptable recovery times, limit the DIO design to a single DIO main ring without sub-rings.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.

Section 6.3
Hot Standby PAC, Local Rack BMENOS0300 Topologies

BMENOS0300 in Local Rack of Hot Standby PACs in support of DIO Ring

Hot Standby DIO Ring

In this Hot Standby topology, two BMENOS0300 network option switches are used in place of BMENOC03•1 modules to reduce cost. The network ports of two BMENOS0300 network option switch modules – one in the primary and one in the standby – are configured for a **HSBY/DIO**.



1 Primary Hot Standby PAC
2 Standby Hot Standby PAC

3 Distributed equipment
4 ConneXium dual ring Ethernet managed switches
5 SCADA server
6 Engineering workstation with dual network interface cards
7 Ethernet DIO ring
8 Hot Standby communication link
9 Ethernet control network
X BMENOC03•1 communication module with backplane port disabled

NOTE:

- In this design, confirm that the backplane port of each BMENOC03•1 communication module is disabled. The backplane port is disabled by default.

⚠ CAUTION
RISK OF BROADCAST STORM Do not enable the Ethernet backplane ports of the primary and standby BMENOC03•1 communication modules. Enabling these ports can result in the occurrence of a broadcast storm, which can prevent the Hot Standby network from transporting intended network communications. **Failure to follow these instructions can result in injury or equipment damage.**

- To achieve acceptable recovery times, limit each DIO design to a single DIO main ring without sub-rings.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the two BMENOS0300 modules.
- When you set the BMENOS0300 network rotary switch to **HSBY/DIO**, assign RSTP bridge identifier priority values greater than 4096 (decimal) to other distributed equipment Ethernet devices in the DIO ring. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **HSBY/DIO**.

Section 6.4
Hot Standby PAC, RIO Drop Topologies for the BMENOS0300

What Is in This Section?

This section contains the following topics:

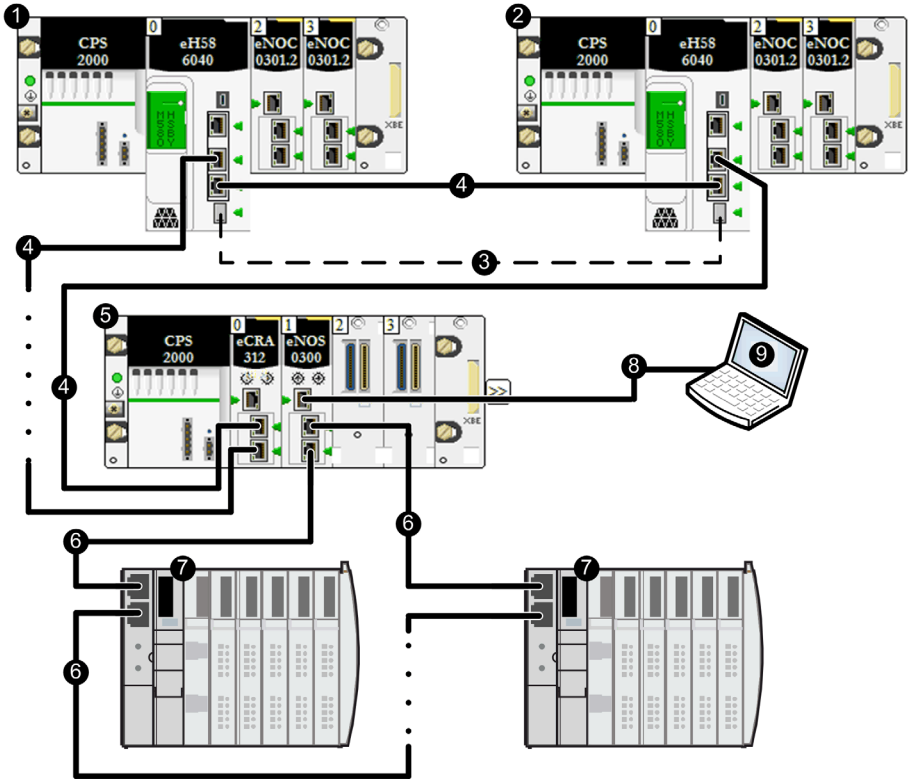
Topic	Page
BMENOS0300 in RIO Drop of Hot Standby PACs in support of DIO Ring	48
BMENOS0300 in RIO Drop of Hot Standby PACs in support of DIO Clouds	50
BMENOS0300 in RIO Drop of Hot Standby PACs in support of RIO Sub-Ring	52
Two BMENOS0300 Modules in RIO Drop of Hot Standby PACs Supporting RIO Sub-Ring and DIO Ring	54

BMENOS0300 in RIO Drop of Hot Standby PACs in support of DIO Ring

Simple DIO Ring

In this Hot Standby topology, the BMENOS0300 network option switch is used in place of a BMENOC03•1 to reduce cost. The BMENOS0300 network option switch is located in an RIO drop. Its ports are used as follows:

- The two network ports are configured for a **DIO Ring**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Primary Hot Standby PAC
2 Standby Hot Standby PAC
3 Hot Standby communication link
4 Ethernet RIO main ring
5 (e)X80 RIO drop
6 Ethernet DIO ring

7 Distributed equipment
8 daisy chain
9 Engineering tool resident on PC

NOTE:

- To achieve acceptable recovery times for the DIO ring, limit its design to a single DIO main ring without sub-rings.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.

BMENOS0300 in RIO Drop of Hot Standby PACs in support of DIO Clouds

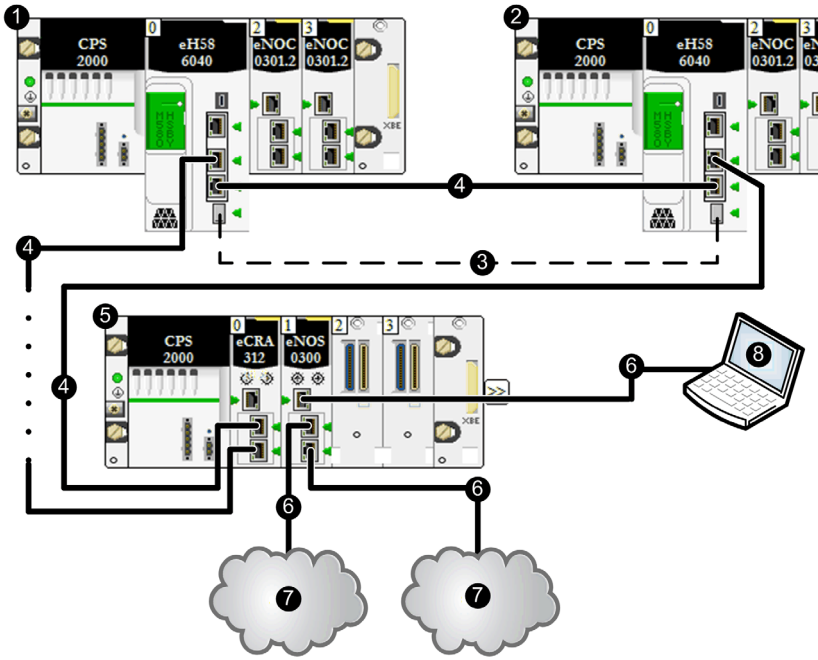
RIO Main Ring with DIO Clouds

In this Hot Standby topology, the BMENOS0300 network option switch is used in an RIO drop. It is used instead of a BMENOC03•1 module to reduce cost. Its ports are used as follows:

- The two network ports are configured as **DIO Ports**.
- The service port is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.

⚠ CAUTION
RISK OF BROADCAST STORM Do not connect an Ethernet device in one DIO cloud to an Ethernet device in another DIO cloud. Connecting these devices can result in the occurrence of a broadcast storm, which can overload the network and delay or prevent intended network communications. **Failure to follow these instructions can result in injury or equipment damage.**

NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Primary Hot Standby PAC
2 Standby Hot Standby PAC
3 Hot Standby communication link

4 Ethernet RIO main ring
5 (e)X80 RIO drop
6 Ethernet non-looping daisy chain
7 DIO cloud
8 Engineering tool resident on PC

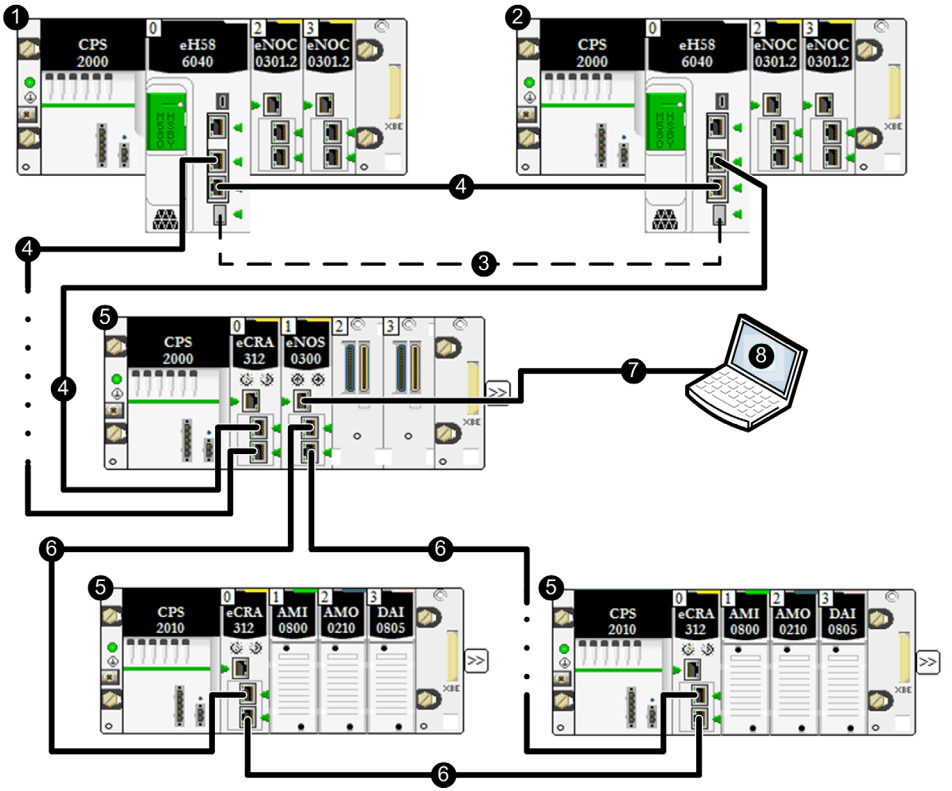
NOTE: Because RSTP is not enabled, there is no RSTP-based maximum device limit.

BMENOS0300 in RIO Drop of Hot Standby PACs in support of RIO Sub-Ring

RIO Main Ring with RIO Sub-Ring

In this Hot Standby topology, the BMENOS0300 network option switch resides in an RIO drop. It is used in place of an external dual ring switch (DRS) to reduce cost. Its ports are configured as follows:

- The two network ports are configured for an **RIO Ring**.
- The service port supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.



1 Primary Hot Standby PAC
2 Standby Hot Standby PAC
3 Hot Standby communication link
4 Ethernet RIO main ring
5 (e)X80 RIO drop
6 Ethernet RIO sub-ring
7 Ethernet daisy chain
8 Engineering tool resident on PC

NOTE:
To achieve desired RSTP-based network response time targets, the maximum combined size of the RIO main ring plus a single RIO sub-ring is 32 switched devices, including:

- both Hot Standby CPUs
- the remote adapter (BMECRA312•0)
- the embedded switch in the Ethernet rack on which the BMENOS0300 is installed
- the BMENOS0300
- up to 27 additional switched devices

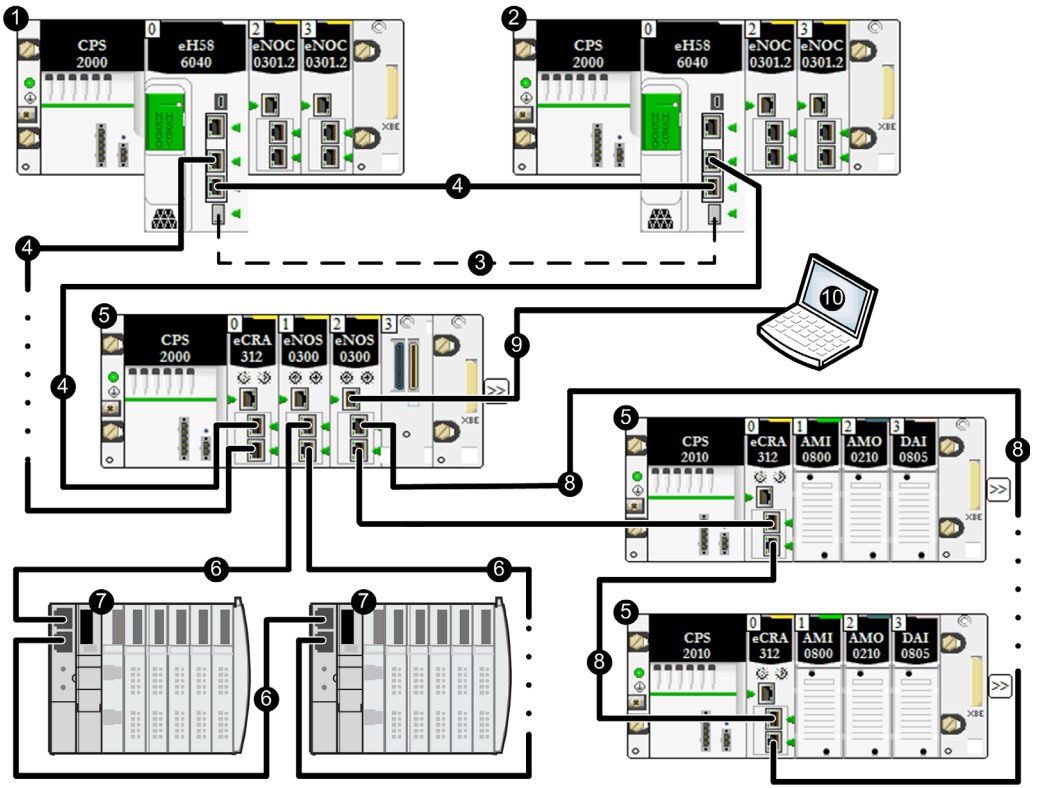
NOTE: When you set the BMENOS0300 network rotary switch to **RIO Ring**, configure RSTP bridge identifier priority settings for other devices in the RIO sub-ring to values greater than 0. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **RIO Ring**.

Two BMENOS0300 Modules in RIO Drop of Hot Standby PACs Supporting RIO Sub-Ring and DIO Ring

RIO Sub-Ring and DIO Ring

In this Hot Standby topology, two BMENOS0300 network option switch modules are used in place of external dual ring switches (DRSs) to reduce cost. These modules reside in an RIO drop. The ports of these modules are used as follows:

- Two network ports of one BMENOS0300 module are configured for an **RIO Ring**, and support an RIO sub-ring.
- The service port of the same module is configured as an **Access** port, and supports an engineering tool (for example, Unity Pro) resident on a PC.
  NOTE: Instead of a PC, a DIO cloud could be connected to the service port.
- The two network ports of the other BMENOS0300 module are configured for a **DIO Ring**.



1 Primary Hot Standby PAC
2 Standby Hot Standby PAC
3 Hot Standby communication link
4 Ethernet main RIO ring

5 (e)X80 RIO drop
6 Ethernet DIO ring
7 Distributed equipment
8 Ethernet RIO sub-ring
9 Ethernet daisy chain
10 Engineering tool resident on PC

NOTE:

- To achieve desired RSTP-based network response time targets, the longest path that an Ethernet RIO packet would have to travel in the case of a cable break is 32 switched devices, including:
  - both Hot Standby CPUs
  - the remote adapter (BMECRA312•0)
  - the embedded switch in the Ethernet rack on which the BMENOS0300 modules are installed
  - the BMENOS0300 module
- When you set the BMENOS0300 network rotary switch to **RIO Ring**, configure the RSTP bridge identifier priority settings of other devices in the RIO sub-ring to values greater than 0. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **RIO Ring**.
- To achieve acceptable recovery times, limit the DIO design to a single DIO main ring without sub-rings.
- When you set the BMENOS0300 network rotary switch to **DIO Ring**, configure a distributed equipment Ethernet device to be the RSTP root bridge. Refer to the topic Device Rotary Switch Positions *(see page 25)* for information about the effects of selecting **DIO Ring**.
- RSTP limits the size of the DIO ring to a maximum of 40 switched devices, including the BMENOS0300 module.

Chapter 7
Ethernet Services

Ethernet Services

Introduction

The BMENOS0300 network option switch provides the following Ethernet services:

- HTTP web page server
- SNMP network management agent
- QoS message priority tagging
- DHCP client for IP address assignment
- FTP server, for firmware upgrade

As implemented in the BMENOS0300 network switch module, all services are pre-configured, and do not require custom configuration.

HTTP Web Page Server

The BMENOS0300 network switch module includes a Hypertext Transfer Protocol (HTTP) server. Use the server to access the module's web pages *(see page 64)* for the purpose of monitoring and diagnosing the module. All web pages are read-only. The server provides easy access to the module from standard internet browsers.

The HTTP web pages are accessed using the IP address *(see page 26)* of the module.

SNMP Network Management Agent

The BMENOS0300 network switch module includes an SNMP v1 agent. An SNMP agent is a software component running on the module that allows access to the module's diagnostic and management information via the SNMP service. SNMP browsers, network management software, and other tools typically use SNMP to access this data. The SNMP agent is read-only.

The SNMP agent is accessed using the IP address *(see page 26)* of the module.

QoS Message Priority Tagging

The BMENOS0300 network switch module supports the OSI layer 3 Quality of Service (QoS) standard defined in RFC-2475, by accepting and re-transmitting Ethernet packets that contain a differentiated services code point (DSCP). The QoS tags are added to Ethernet packets by other devices that support QoS configuration.

DHCP Client

The BMENOS0300 network option switch includes a DHCP client. The purpose of the DHCP client is to receive, from a DHCP server, an IP address for Ethernet network communications. When powered up, the module sends a request, based on the module's device name, to the DHCP server resident in the CPU. The DHCP server delivers an IP address to the module, if the DHCP server has been configured to do so. If not, the module assigns itself an IP address based on its MAC Address.

FTP Server

The BMENOS0300 network option switch includes an FTP server. The FTP server is continuously enabled.

The FTP server is accessed using the IP address *(see page 26)* of the module.

Chapter 8
Cyber Security

Cyber Security

Introduction

The BMENOS0300 module is an inexpensive, easy-to-use network switch that is configured exclusively by means of the two rotary switches. In addition, it receives its IP address from a DHCP server, based on the auto-generated device name *(see page 28)*. As a simple device, the BMENOS0300 has very limited functionality. Its core functionality, switching, is configured using rotary switches. Diagnostics are provided by IP based Ethernet services.

This topic shows you how to provide CPU-based Ethernet services and switching functionality to the BMENOS0300 module.

Access Control

Because the BMENOS0300 is a switching module, there is no need for it to provide ACL protection to IP based services. All Ethernet packets are pass through. Hence protection can be provided for the connected end devices (such as the CPU and the Ethernet I/O adapter module).

The BMENOS0300 module accepts Ethernet packets sent to its Ethernet ports from connected Ethernet devices. If you wish to limit the inflow of Ethernet packets into your application, you can enable **Access Control** in the **Security** tab of the M580 CPU module DTM. Access Control restricts device access to the CPU in its role as a server. You can add the IP addresses of the devices that you want to communicate with the CPU to the list of Authorized Addresses:

- By default, the IP address of the CPU's embedded Ethernet I/O scanner service with subnet set to Yes allows any device in the subnet to communicate with the CPU through EtherNet/IP or Modbus TCP.
- Add the IP address of any client device that may send a request to the CPU's Ethernet I/O scanner service, which, in this case, acts as a Modbus TCP or EtherNet/IP server.
- Add the IP address of your maintenance PC to communicate with the PAC through the CPU Ethernet I/O scanner service via Unity Pro to configure and diagnose your application.

NOTE: The subnet in the IP Address column can be the subnet itself or any IP address inside the subnet. If you select **Yes** for a subnet that does not have a subnet mask, a pop-up window states that the screen cannot be validated because of a detected error.

Disabling Ethernet Services

Ethernet access to the BMENOS0300 module is not enabled until it is served an IP address from a DHCP server.

If you wish to disable Ethernet services for the module, do not assign it an IP address. In this configuration, the module continues to operate as an Ethernet switch, but does not initiate its Ethernet services.

When the BMENOS0300 receives an IP address, the information that can be accessed via its DDDT *(see page 73)* is read-only diagnostic data, which necessarily presents only a limited security concern. If the BMENOS0300 stops functioning properly as the result of a cyber attack, the module enters a reduced functionality operating mode, in which its switching function is disabled. This response limits the likelihood of the attack affecting other devices in the Ethernet network.

Disabling Switch Ports

You can disable the service port by setting the **SERVICE** rotary switch to the **DISABLED** position. You cannot disable the two network ports or the backplane port. However, as noted above, you can configure access control *(see page 59)* for the M580 CPU in your application.

Chapter 9
Diagnostics

Overview

This chapter presents diagnostic tools for the BMENOS0300 network switch module.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
9.1	BMENOS0300 Diagnostic LEDs	62
9.2	BMENOS0300 Module Embedded Web Pages	64
9.3	BMENOS0300 DDDT	73

Section 9.1
BMENOS0300 Diagnostic LEDs

LED Diagnostics for the BMENOS0300 Module

Overview

The BMENOS0300 module includes diagnostic LEDs:

- In a panel at the top of the module front face.
- Next to each RJ45 Ethernet connector.

LED Panel Display

A 4-LED display is located on the front panel of the module:



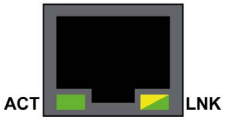
The BMENOS0300 panel LEDs indicate the following states:

LED	Color	State	Description
RUN	Green	On	The module is operating normally.
		Off	The module is powered off or is not configured.
		Flashing	The module is in STOP mode, self-test mode, duplicate IP mode, or there is no Ethernet link.
ERR	Red	Off	The module is operating normally; no error is detected.
		Flashing	The module is not configured, a configuration is in progress, or a backplane communication error is detected.
		On	An error is detected.
MS	Green	On	The module is operating normally.
		Flashing	The module has not been configured.
	Green / Red	Off	Power is not supplied to the module.
	Red	On	A non-recoverable fault is detected.
		Flashing	A recoverable fault is detected.

LED	Color	State	Description
NS	Green	On	The module has established at least one CIP connection.
		Flashing	An IP address has been assigned, but the no CIP connections are established.
	Green / Red	Off	No IP address has been assigned to the module.
	Red	On	A duplicate IP address is detected.
		Flashing	One or more CIP connections, for which the module is a target, have timed out.

Ethernet Port Connector LEDs

Each Ethernet RJ45 connector presents a pair of LED indicators:



The Ethernet connector LEDs indicate the following states:

LED	Color	State	Description
ACT	Green	Flashing	Data is being transmitted over the link.
		Off	No transmission activity is occurring.
LNK	Green	On	Link speed = 100 Mbit/s.
	Yellow	On	Link speed = 10 Mbit/s.
	Green / Yellow	Off	No link is established.

Section 9.2
BMENOS0300 Module Embedded Web Pages

Introduction

The BMENOS0300 network switch module includes an HTTP server. The server transmits web pages for the purpose of monitoring and diagnosing the module. The server provides easy access to the module from standard internet browsers.

Use the module IP address *(see page 26)* to access the web pages.

What Is in This Section?

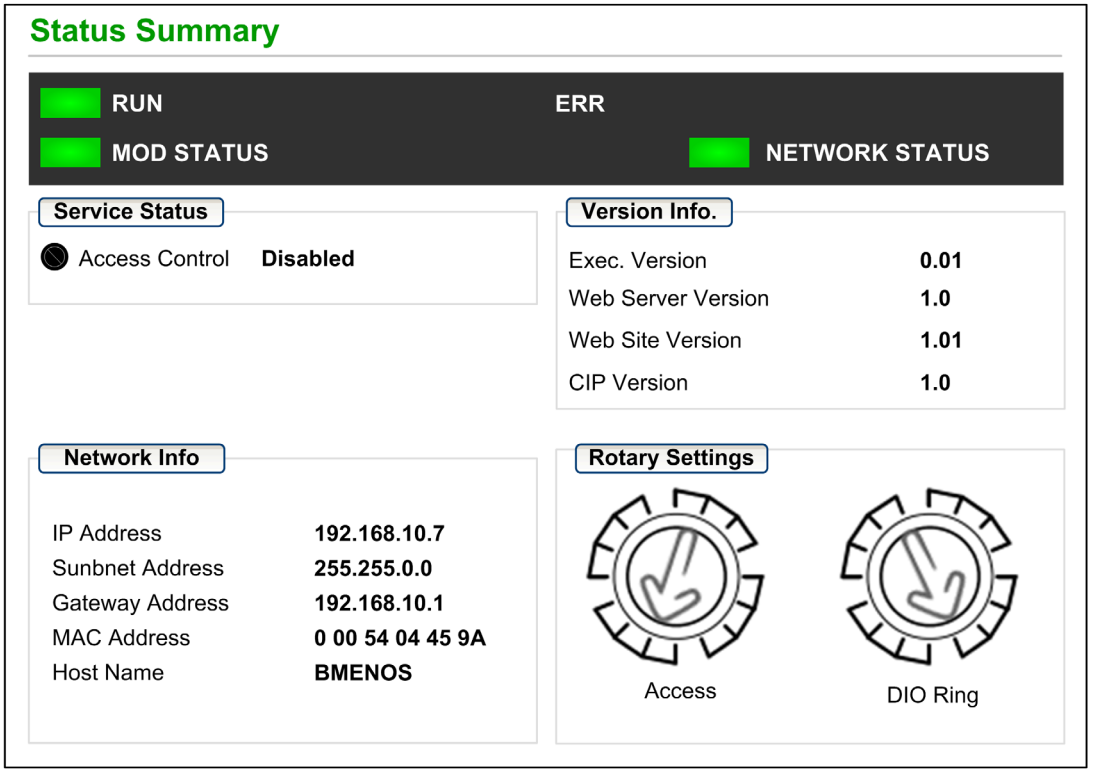
This section contains the following topics:

Topic	Page
Status Summary	65
Port Statistics	67
QoS	69
Redundancy	71

Status Summary

Open the Page

Access the **Status Summary** page from the **Diagnostics** tab (**Menu → Module → Summary**):



NOTE: This page is updated every 5 seconds.

Diagnostic Information

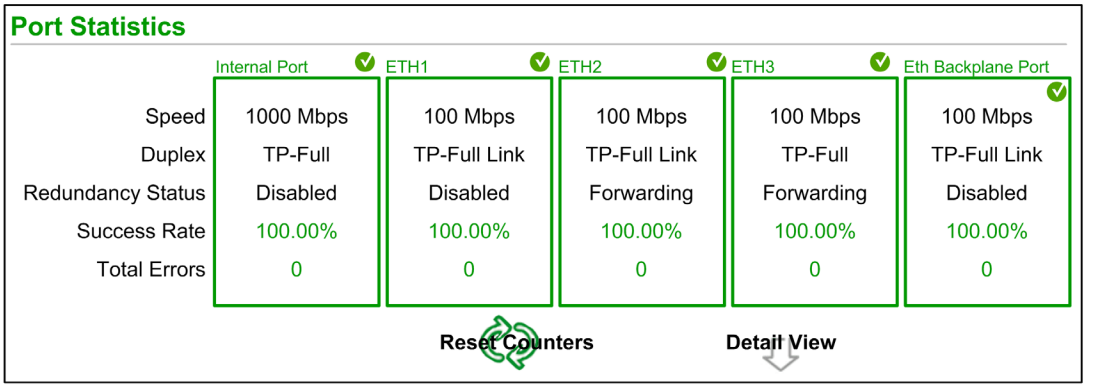
The objects on this page provide status information:

Parameters	Description	
LEDs	The LED panel includes the following LED indicators: ● RUN ● ERR ● MOD STATUS ● NETWORK STATUS **NOTE:** The LEDs are described in the description of the LED panel. *(see page 62)*	
Service Status	green	The available service is operational and running.
	red	An error is detected in an available service.
	black	The available service is not present or not configured.
Version Info.	This field describes the software versions that are running on the module.	
Network Info.	This field contains network and hardware address information and connectivity that corresponds to the module.	
Rotary Settings	This field describes the position of the two rotary switches *(see page 24)* on the front of the module: ● The left switch setting describes the configuration of the service port. ● The right switch setting describes the configuration of the two network ports.	

Port Statistics

Open the Page

Access the **Port Statistics** page from the **Diagnostics** tab (**Menu → Module → Port Statistics**):



NOTE: This page is updated every 5 seconds. Click **Reset Counters** to reset all dynamic counters to 0.

Diagnostic Information

This page shows the statistics for each port on the module. This information is associated with the configuration of the Service port (ETH 1) and the Device network ports (ETH 2 and ETH 3) *(see page 24)*.

The frame color indicates the port activity:

- *green*: active
- *gray*: inactive
- *yellow*: recoverable error detected
- *red*: non-recoverable error detected

Expanded View

Click **Detail View** to see more statistics:

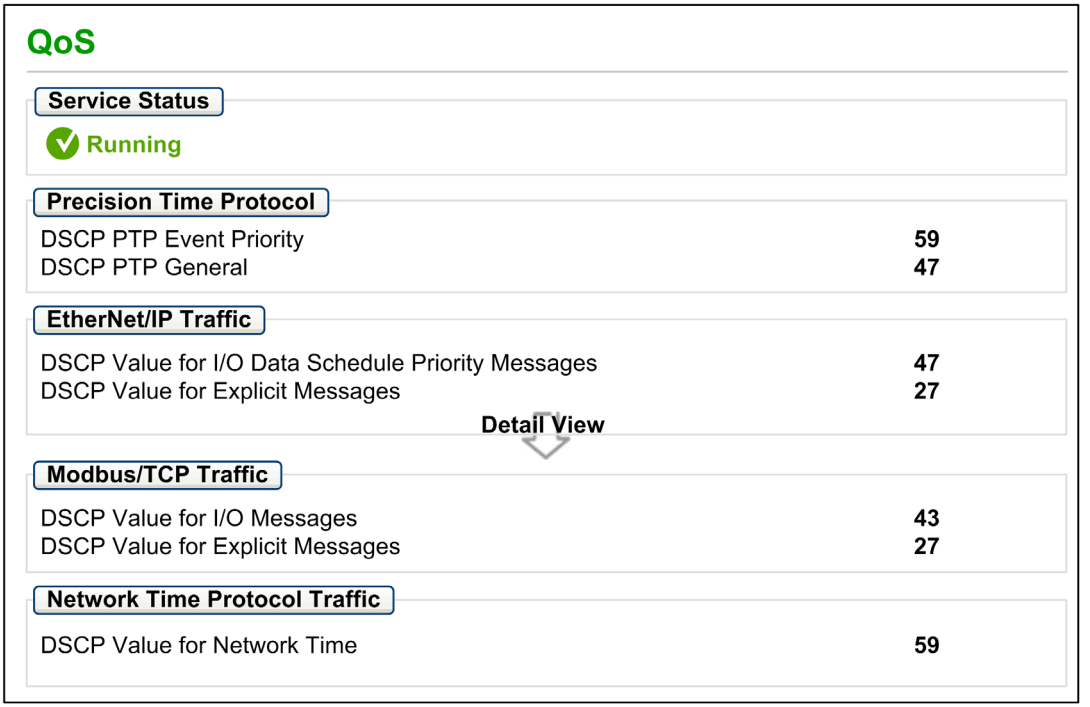
Statistic	Description
Frames Transmitted	Number of frames successfully transmitted.
Frames Received	Number of frames received.
Excessive Collisions	Number of excessive Ethernet collisions.
Late Collisions	Number of late Ethernet collisions.
CRC Errors	Number of detected cyclic redundancy check (CRC) errors.

Statistic	Description
Bytes Received	Number of bytes received.
Inbound Packet Errors	Number of detected inbound packet errors.
Inbound Packets Discarded	Number of inbound packets discarded.
Bytes Transmitted	Number of bytes transmitted.
Outbound Packet Errors	Number of detected outbound packet errors.
Outbound Packets Discarded	Number of outbound packets discarded.
Carrier Sense Errors	Number of detected carrier sense errors. A carrier sense error is detected when a port tries to transmit a frame, but cannot do so because no carrier is detected.
FCS Errors	Number of detected frame check sequence (FCS) errors. An FCS error is detected when a frame is corrupted during transmission as indicated by its checksum value.
Alignment Errors	The number of byte alignment errors that have been detected. A byte alignment occurs when the number of bits in a frame is not divisible by 8. An alignment error also triggers an FCS error.
Internal MAC Trans. Errors	The number of detected transmit errors that are not late collisions, excessive collisions, or CRC errors.
Internal MAC Rec. Errors	The number of detected receive errors that are not late collisions, excessive collisions, or CRC errors.
SQE Test Errors	The number of detected signal quality error (SQE) instances. Some Ethernet transceivers use an SQE heartbeat to indicate it is connected to a host interface. This detected error indicates that a transceiver has no heartbeat. Note that not all transceivers produce a heartbeat.

QoS

Open the Page

Access the **QoS** (quality of service) page from the **Diagnostics** tab (**Menu → Services → QoS**):



NOTE:

- QoS values are automatically set when you select positions for the rotary switches *(see page 24)*.
- Click **Detail View** to expand the list of parameters.
- This page is updated every 5 seconds.

Service Status

This table shows the possible states for the **Service Status**:

Status	Description
Running	The service is correctly configured and running.
Disabled	The service is disabled.
Uknown	The status of the service is not known.

Diagnostic Information

QoS causes the module to add a differentiated services code point (DSCP) tag to each Ethernet packet it transmits, thereby indicating the priority of that packet:

Field	Parameter	Description
Precision Time Protocol	DSCP PTP Event Priority	Point-to-point time synchronization.
	DSCP PTP General	Point-to-point general.
EtherNet/IP Traffic	DSCP Value for I/O Data Scheduled Priority Messages	Configure the priority levels to prioritize the management of data packets.
	DSCP Value for Explicit Messages	
Modbus/TCP Traffic	DSCP Value for I/O Messages	**NOTE:** We recommend that you use a larger timeout value for explicit messaging connections and a smaller timeout value for implicit messaging connections. The specific values that you employ depend on your application requirements.
	DSCP Value for Explicit Messages	
Network Time Protocol Traffic	DSCP Value for Network Time	—

Considerations

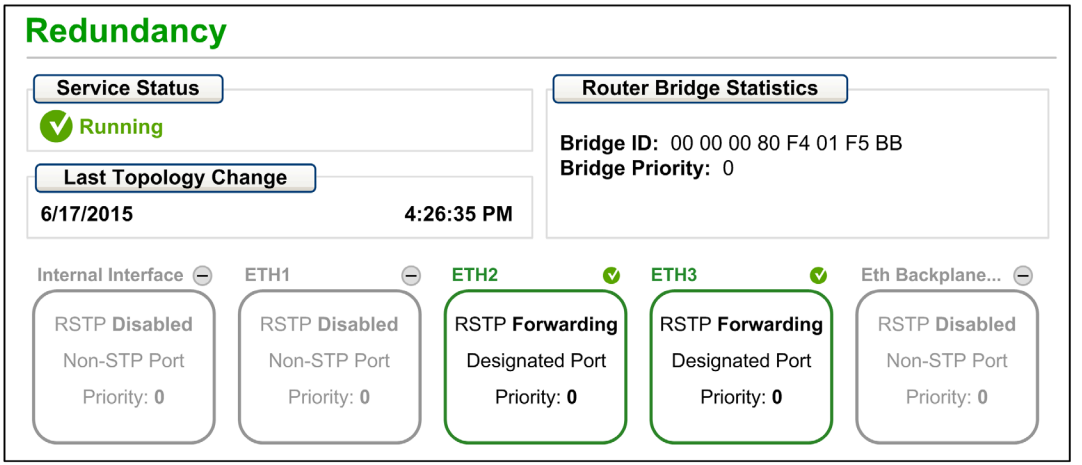
Take measures to effectively implement QoS settings in your Ethernet network:

- Use only network switches that support QoS.
- Apply the same DSCP values to all network devices and switches.
- Use switches that apply a consistent set of rules for handling the different DSCP values when transmitting and receiving Ethernet packets.

Redundancy

Open the Page

Access the **Redundancy** page on the **Diagnostic** tab (**Menu → Services → Redundancy**):



NOTE:

- When redundancy is enabled in the configuration, both port ETH2 and port ETH3 are designated as RSTP ports. No other module port can be configured to support the RSTP protocol.
- This page is updated every 5 seconds.

Diagnostic Information

This page displays RSTP-related data:

Field	Description	
Service Status	**Running**	The RSTP bridge on the corresponding CPU is properly configured and running.
	Disabled	The RSTP bridge on the corresponding CPU is disabled.
	Unknown	The status of the RSTP bridge on the corresponding CPU is not known.
Last Topology Change	These values represent the date and time that the last topology change was received for the corresponding **Bridge ID**.	
Redundancy Status	green	The designated Ethernet port is learning or formatting information.
	yellow	The designated Ethernet port is discarding information.
	gray	RSTP is disabled for the designated Ethernet port.

Field	Description	
Router Bridge Statistics	**Bridge ID**	This unique bridge identifier is the concatenation of the bridge RSTP priority and the MAC address.
	Bridge Priority	The RSTP priority of the port.

Section 9.3
BMENOS0300 DDDT

BMENOS0300 DDDT

Introduction

BMENOS0300 network option switch includes the following 16-Byte DDDT, which you can interrogate to diagnose the status of the module.

Byte	Bit Offset	Parameter	Description
0	ETH_STATUS		
	0.0	PORT1_LINK[1]	0: Link down / 1: Link up
	0.1	PORT2_LINK[1]	0: Link down / 1: Link up
	0.2	PORT3_LINK[1]	0: Link down / 1: Link up
	0.3	PORT4_LINK[1]	0: Link down / 1: Link up
	0.4	<Reserved>	–
	0.5	REDUNDANCY_STATUS	0: Ring broken or status unknown / 1: Ring intact
	0.6	<Reserved>	–
	0.7	GLOBAL_STATUS	0: Non-normal operation / 1: Normal operation
1	SERVICE_STATUS		
	1.0	RSTP_SERVICE	0: Non-normal operation / 1: Normal operation
	1.1	PORT502_SERVICE	0: Non-normal operation / 1: Normal operation
	1.2	SNMP_SERVICE	0: Non-normal operation / 1: Normal operation
	1.3	WEB_SERVER	0: Non-normal operation / 1: Normal operation
	1.4	ETH_BKP_FAILURE	0: Detected error / 1: OK
	1.5	ETH_BKP_ERROR	0: Detected error / 1: OK
	1.6	FIRMWARE_UPGRADE	0: Non-normal operation / 1: Normal operation
	1.7	EIP_ADAPTER	0: Non-normal operation / 1: Normal operation
2	ETH_PORT_1_2_STATUS		
	2.0...2.1	PORT1_FUNCTION	0:Disabled / 1: Access / 2: Mirror / 3: Network
	2.2...2.3	PORT1_RSTP_ROLE	0:Alternate / 1: Backup / 2: Designated / 3: Root
	2.4...2.5	PORT2_FUNCTION	0:Disabled / 1: Access / 2: Mirror / 3: Network
	2.6...2.7	PORT2_RSTP_ROLE	0:Alternate / 1: Backup / 2: Designated / 3: Root
1. Port number references are: Port 1: ETH1 / Port 2: ETH2 / Port 3: ETH 3 / Port 4: Backplane port			

Byte	Bit Offset	Parameter	Description
3	ETH_PORT_3_BKP_STATUS		
	3.0...3.1	PORT3_FUNCTION	0:Disabled / 1: Access / 2: Mirror / 3: Network
	3.2...3.3	PORT3_RSTP_ROLE	0:Alternate / 1: Backup / 2: Designated / 3: Root
	3.4...3.5	PORT4_FUNCTION	0:Disabled / 1: Access / 2: Mirror / 3: Network
	3.6...3.7	PORT4_RSTP_ROLE	0:Alternate / 1: Backup / 2: Designated / 3: Root
4...5	MODULE_PORT_FUNCTIONS		
	4.0...5.3	<Reserved>	–
	5.4...5.5	SERVICE_PORT_FUNCTION	0:Disabled / 1: Access / 2: Mirror / 3: Unused
	5.6...5.7	NETWORK_PORT_FUNCTION	0: RIO Ring / 1: DIO Ring / 2: DIO Ports / 3: DIO Root
6...7	MODULE_STATUS_FLAGS		
	6.0...7.6	<Reserved>	–
	7.7	POTENTIAL_STORM_DETECT	0: Normal / 1: Potential storm
8...11	POTENTIAL_STORM_COUNTER		Potential storm counter (integer)
12...15	<Reserved>		–
1. Port number references are: Port 1: ETH1 / Port 2: ETH2 / Port 3: ETH 3 / Port 4: Backplane port			

Establishing a Connection to the DDDT via EtherNet/IP

You can establish a read-only connection to the BMENOS0300 DDDT using either of the following EtherNet/IP connection types:

- Transport class 1: Implicit scheduled data transfer (I/O) connection (UDP/IP)
- Transport class 3: Explicit unscheduled messaging via the CIP Assembly Object (TCP/IP)

Establishing an Implicit Messaging Connection

You can establish an implicit messaging connection to the module using the following connection configuration information:

- Connection transport class: 01 (decimal)
- Connection type: O->T Exclusive Owner
- Instance ID: 101 (decimal)
- Data size: 16 bytes (decimal)

Sending an Explicit Message

You can use the CIP Assembly object of the BMENOS0300 module to send a one-time read-only explicit message to the module.

NOTE: You can send an explicit message to the Assembly object only when no other connections have been established that read from or write to this object.

The Assembly object includes the following attributes and services:

- Connection transport class: 04 (decimal)
- Instance ID: 101 (decimal)
- Attribute ID: 03 (decimal)
- Service ID: 14 (decimal) / 0E (hex): Get_Attribute_Single

Reading the Response

The response of either an implicit or an explicit message consists of 16 hexadecimal byte values, with the least significant byte (0) on the left, and the most significant byte (15) on the right.

Glossary

!

%I

According to the CEI standard, %I indicates a language object of type discrete IN.

%IW

According to the CEI standard, %IW indicates a language object of type analog IN.

%M

According to the CEI standard, %M indicates a language object of type memory bit.

%MW

According to the CEI standard, %MW indicates a language object of type memory word.

%Q

According to the CEI standard, %Q indicates a language object of type discrete OUT.

%QW

According to the CEI standard, %QW indicates a language object of type analog OUT.

%SW

According to the CEI standard, %SW indicates a language object of type system word.

A

adapter

An adapter is the target of real-time I/O data connection requests from scanners. It cannot send or receive real-time I/O data unless it is configured to do so by a scanner, and it does not store or originate the data communications parameters necessary to establish the connection. An adapter accepts explicit message requests (connected and unconnected) from other devices.

advanced mode

In Unity Pro, advanced mode is a selection that displays expert-level configuration properties that help define Ethernet connections. Because these properties should be edited only by people with a good understanding of EtherNet/IP communication protocols, they can be hidden or displayed, depending upon the qualifications of the specific user.

architecture

Architecture describes a framework for the specification of a network that is constructed of these components:

- physical components and their functional organization and configuration
- operational principles and procedures
- data formats used in its operation

ARRAY

An `ARRAY` is a table containing elements of a single type. This is the syntax: `ARRAY [<limits>] OF <Type>`

Example: `ARRAY [1..2] OF BOOL` is a one-dimensional table with two elements of type `BOOL`.

`ARRAY [1..10, 1..20] OF INT` is a two-dimensional table with 10x20 elements of type `INT`.

ART

(*application response time*) The time a CPU application takes to react to a given input. ART is measured from the time a physical signal in the CPU turns on and triggers a write command until the remote output turns on to signify that the data has been received.

AUX

An (AUX) task is an optional, periodic processor task that is run through its programming software. The AUX task is used to execute a part of the application requiring a low priority. This task is executed only if the MAST and FAST tasks have nothing to execute. The AUX task has two sections:

- IN: Inputs are copied to the IN section before execution of the AUX task.
- OUT: Outputs are copied to the OUT section after execution of the AUX task.

B

BCD

(*binary-coded decimal*) Binary encoding of decimal numbers.

BOOL

(*boolean type*) This is the basic data type in computing. A `BOOL` variable can have either of these values: 0 (`FALSE`) or 1 (`TRUE`).

A bit extracted from a word is of type `BOOL`, for example: `%MW10.4`.

BOOTP

(*bootstrap protocol*) A UDP network protocol that can be used by a network client to automatically obtain an IP address from a server. The client identifies itself to the server using its MAC address. The server, which maintains a pre-configured table of client device MAC addresses and associated IP addresses, sends the client its defined IP address. The BOOTP service utilizes UDP ports 67 and 68.

broadcast

A message sent to all devices in a broadcast domain.

C

CCOTF

(*change configuration on the fly*) A feature of Unity Pro that allows a module hardware change in the system configuration while the system is operating. This change does not impact active operations.

CIP™

(*common industrial protocol*) A comprehensive suite of messages and services for the collection of manufacturing automation applications (control, safety, synchronization, motion, configuration and information). CIP allows users to integrate these manufacturing applications with enterprise-level Ethernet networks and the internet. CIP is the core protocol of EtherNet/IP.

class 1 connection

A CIP transport class 1 connection used for I/O data transmission via implicit messaging between EtherNet/IP devices.

class 3 connection

A CIP transport class 3 connection used for explicit messaging between EtherNet/IP devices.

connected messaging

In EtherNet/IP, connected messaging uses a CIP connection for communication. A connected message is a logical relationship between two or more application objects on different nodes. The connection establishes a virtual circuit in advance for a particular purpose, such as frequent explicit messages or real-time I/O data transfers.

connection

A virtual circuit between two or more network devices, created prior to the transmission of data. After a connection is established, a series of data is transmitted over the same communication path, without the need to include routing information, including source and destination address, with each piece of data.

connection originator

The EtherNet/IP network node that initiates a connection request for I/O data transfer or explicit messaging.

connectionless

Describes communication between two network devices, whereby data is sent without prior arrangement between the two devices. Each piece of transmitted data also includes routing information, including source and destination address.

control network

An Ethernet-based network containing PACs, SCADA systems, an NTP server, PCs, AMS, switches, etc. Two kinds of topologies are supported:

- flat: All modules and devices in this network belong to same subnet.
- 2 levels: The network is split into an operation network and an inter-controller network. These two networks can be physically independent, but are generally linked by a routing device.

CPU

(*central processing unit*) The CPU, also known as the processor or controller, is the brain of an industrial manufacturing process. It automates a process as opposed to relay control systems. CPUs are computers suited to survive the harsh conditions of an industrial environment.

D

DDT

(*derived data type*) A derived data type is a set of elements with the same type (`ARRAY`) or with different types (structure).

determinism

For a defined application and architecture, you can predict that the delay between an event (change of value of an input) and the corresponding change of a controller output is a finite time *t*, smaller than the deadline required by your process.

Device DDT (DDDT)

A Device DDT is a DDT predefined by the manufacturer and not modifiable by user. It contains the I/O language elements of an I/O module.

device network

An Ethernet-based network within a remote I/O network that contains both remote I/O and distributed I/O devices. Devices connected on this network follow specific rules to allow remote I/O determinism.

device network

An Ethernet-based network within an RIO network that contains both RIO and distributed equipment. Devices connected on this network follow specific rules to allow RIO determinism.

DFB

(*derived function block*) DFB types are function blocks that can be defined by the user in ST, IL, LD or FBD language.

Using these DFB types in an application makes it possible to:

- simplify the design and entry of the program
- make the program easier to read
- make it easier to debug
- reduce the amount of code generated

DHCP

(*dynamic host configuration protocol*) An extension of the BOOTP communications protocol that provides for the automatic assignment of IP addressing settings, including IP address, subnet mask, gateway IP address, and DNS server names. DHCP does not require the maintenance of a table identifying each network device. The client identifies itself to the DHCP server using either its MAC address, or a uniquely assigned device identifier. The DHCP service utilizes UDP ports 67 and 68.

DIO

(*distributed I/O*) Legacy term for distributed equipment. DRSs use DIO ports to connect distributed equipment.

DIO cloud

A group of distributed equipment that is not required to support RSTP. DIO clouds require only a single (non-ring) copper wire connection. They can be connected to some of the copper ports on DRSs, or they can be connected directly to the CPU or Ethernet communications modules in the *local rack*. DIO clouds **cannot** be connected to *sub-rings*.

DIO network

A network containing distributed equipment, in which I/O scanning is performed by a CPU with DIO scanner service on the local rack. DIO network traffic is delivered after RIO traffic, which takes priority in an RIO network.

distributed equipment

Any Ethernet device (Schneider Electric device, PC, servers, or third-party devices) that supports exchange with a CPU or other Ethernet I/O scanner service.

DNS

(*domain name server/service*) A service that translates an alpha-numeric domain name into an IP address, the unique identifier of a device on the network.

domain name

An alpha-numeric string that identifies a device on the internet, and which appears as the primary component of a web site's uniform resource locator (URL). For example, the domain name *schneider-electric.com* is the primary component of the URL *www.schneider-electric.com*.

Each domain name is assigned as part of the domain name system, and is associated with an IP address.

Also called a host name.

DRS

(*dual-ring switch*) A ConneXium extended managed switch that has been configured to operate on an Ethernet network. Predefined configuration files are provided by Schneider Electric to downloaded to a DRS to support the special features of the main ring / sub-ring architecture.

DSCP

(*differentiated service code points*) This 6-bit field is in the header of an IP packet to classify and prioritize traffic.

DST

(*daylight saving time*) DST is also called *summer time* and is a practice consisting of adjusting forward the clock near the start of spring and adjusting it backward near the start of autumn.

DT

(*date and time*) The `DT` type, encoded in BCD in a 64-bit format, contains this information:

- the year encoded in a 16-bit field
- the month encoded in an 8-bit field
- the day encoded in an 8-bit field
- the time encoded in an 8-bit field
- the minutes encoded in an 8-bit field
- the seconds encoded in an 8-bit field

NOTE: The eight least significant bits are not used.

The `DT` type is entered in this format:

`DT#<Year>-<Month>-<Day>-<Hour>:<Minutes>:<Seconds>`

This table shows the upper/lower limits of each field:

Field	Limits	Comment
Year	[1990,2099]	Year
Month	[01,12]	The leading 0 is displayed; it can be omitted during data entry.
Day	[01,31]	For months 01/03/05/07/08/10/12
	[01,30]	For months 04/06/09/11
	[01,29]	For month 02 (leap years)
	[01,28]	For month 02 (non-leap years)
Hour	[00,23]	The leading 0 is displayed; it can be omitted during data entry.
Minute	[00,59]	The leading 0 is displayed; it can be omitted during data entry.
Second	[00,59]	The leading 0 is displayed; it can be omitted during data entry.

DTM

(*device type manager*) A DTM is a device driver running on the host PC. It provides a unified structure for accessing device parameters, configuring and operating the devices, and troubleshooting devices. DTMs can range from a simple graphical user interface (GUI) for setting device parameters to a highly sophisticated application capable of performing complex real-time calculations for diagnosis and maintenance purposes. In the context of a DTM, a device can be a communications module or a remote device on the network.

See FDT.

E

EDS

(*electronic data sheet*) EDS are simple text files that describe the configuration capabilities of a device. EDS files are generated and maintained by the manufacturer of the device.

EF

(*elementary function*) This is a block used in a program which performs a predefined logical function.

A function does not have any information on the internal state. Several calls to the same function using the same input parameters will return the same output values. You will find information on the graphic form of the function call in the [*functional block (instance)*]. Unlike a call to a function block, function calls include only an output which is not named and whose name is identical to that of the function. In FBD, each call is indicated by a unique [number] via the graphic block. This number is managed automatically and cannot be modified.

Position and configure these functions in your program in order to execute your application.

You can also develop other functions using the SDKC development kit.

EFB

(*elementary function block*) This is a block used in a program which performs a predefined logical function.

EFBs have states and internal parameters. Even if the inputs are identical, the output values may differ. For example, a counter has an output indicating that the preselection value has been reached. This output is set to 1 when the current value is equal to the preselection value.

EIO network

Ethernet I/O) An Ethernet-based network that contains three types of devices:

- local rack
- X80 EIO drop, or a Quantum EIO drop (using a BM•CRA312•0 adapter module), or a BMENOS0300 network option switch module
- ConneXium extended dual-ring switch (DRS)

NOTE: Distributed equipment may also participate in an EIO network via connection to DRSs or the service port of X80 EIO adapter modules.

EN

`EN` stands for **EN**able; it is an optional block input. When the `EN` input is enabled, an `ENO` output is set automatically.

If `EN` = 0, the block is not enabled; its internal program is not executed, and `ENO` is set to 0.

If `EN` = 1, the block's internal program is run and `ENO` is set to 1. If a runtime error is detected, `ENO` is set to 0.

If the `EN` input is not connected, it is set automatically to 1.

ENO

`ENO` stands for **E**rror **NO**tification; this is the output associated with the optional input `EN`.

If `ENO` is set to 0 (either because `EN` = 0 or if a runtime error is detected):

- The status of the function block outputs remains the same as it was during the previous scanning cycle that executed correctly.
- The output(s) of the function, as well as the procedures, are set to 0.

Ethernet

A 10 Mb/s, 100 Mb/s, or 1 Gb/s, CSMA/CD, frame-based LAN that can run over copper twisted pair or fiber optic cable, or wireless. The IEEE standard 802.3 defines the rules for configuring a wired Ethernet network; the IEEE standard 802.11 defines the rules for configuring a wireless Ethernet network. Common forms include 10BASE-T, 100BASE-TX, and 1000BASE-T, which can utilize category 5e copper twisted pair cables and RJ45 modular connectors.

Ethernet DIO scanner service

This embedded DIO scanner service of M580 CPUs manages distributed equipment on an M580 device network.

Ethernet I/O scanner service

This embedded Ethernet I/O scanner service of M580 CPUs manages distributed equipment **and** RIO drops on an M580 device network.

EtherNet/IP™

A network communication protocol for industrial automation applications that combines the standard internet transmission protocols of TCP/IP and UDP with the application layer common industrial protocol (CIP) to support both high speed data exchange and industrial control. EtherNet/IP employs electronic data sheets (EDS) to classify each network device and its functionality.

explicit messaging

TCP/IP-based messaging for Modbus TCP and EtherNet/IP. It is used for point-to-point, client/server messages that include both data, typically unscheduled information between a client and a server, and routing information. In EtherNet/IP, explicit messaging is considered class 3 type messaging, and can be connection-based or connectionless.

explicit messaging client

(*explicit messaging client class*) The device class defined by the ODVA for EtherNet/IP nodes that only support explicit messaging as a client. HMI and SCADA systems are common examples of this device class.

F

FAST

A FAST task is an optional, periodic processor task that identifies high priority, multiple scan requests, which is run through its programming software. A FAST task can schedule selected I/O modules to have their logic solved more than once per scan. The FAST task has two sections:

- IN: Inputs are copied to the IN section before execution of the FAST task.
- OUT: Outputs are copied to the OUT section after execution of the FAST task.

FBD

(*function block diagram*) A graphical programming language that works like a flowchart. By adding simple logical blocks (`AND`, `OR`, etc.), each function or function block in the program is represented in this graphical format. For each block, the inputs are on the left and the outputs on the right. Block outputs can be linked to inputs of other blocks in order to create complex expressions.

FDR

(*fast device replacement*) A service that uses configuration software to replace an inoperable product.

FDT

(*field device tool*) The technology that harmonizes communication between field devices and the system host.

FTP

(*file transfer protocol*) A protocol that copies a file from one host to another over a TCP/IP-based network, such as the internet. FTP uses a client-server architecture as well as separate control and data connections between the client and server.

full duplex

The ability of two networked devices to independently and simultaneously communicate with each other in both directions.

function block diagram

See FBD.

G

gateway

A gateway device interconnects two different networks, sometimes through different network protocols. When it connects networks based on different protocols, a gateway converts a datagram from one protocol stack into the other. When used to connect two IP-based networks, a gateway (also called a router) has two separate IP addresses, one on each network.

H

harsh environment

Resistance to hydrocarbons, industrial oils, detergents and solder chips. Relative humidity up to 100%, saline atmosphere, significant temperature variations, operating temperature between -10°C and + 70 °C, or in mobile installations. For hardened (H) devices, the relative humidity is up to 95% and the operating temperature is between -25°C and + 70 °C.

HART

(*highway addressable remote transducer*) A bi-directional communication protocol for sending and receiving digital information across analog wires between a control or monitoring system and smart devices.

HART is the global standard for providing data access between host systems and intelligent field instruments. A host can be any software application from a technician's hand-held device or laptop to a plant's process control, asset management, or other system using any control platform.

high-capacity daisy chain loop

Often referred to as HCDL, a high-capacity daisy chain loop uses dual-ring switches (DRSs) to connect device sub-rings (containing RIO drops or distributed equipment) and/or DIO clouds to the Ethernet RIO network.

HMI

(*human machine interface*) System that allows interaction between a human and a machine.

Hot Standby

A Hot Standby system uses a primary PAC (PLC) and a standby PAC. The two PAC racks have identical hardware and software configurations. The standby PAC monitors the current system status of the primary PAC. If the primary PAC becomes inoperable, high-availability control is maintained when the standby PLC takes control of the system.

HTTP

(*hypertext transfer protocol*) A networking protocol for distributed and collaborative information systems. HTTP is the basis of data communication for the web.

I

I/O scanner

An Ethernet service that continuously polls I/O modules to collect data, status, event, and diagnostics information. This process monitors inputs and controls outputs. This service supports both RIO and DIO logic scanning.

IEC 61131-3

International standard: programmable logic controllers

Part 3: programming languages

IGMP

(internet group management protocol) This internet standard for multicasting allows a host to subscribe to a particular multicast group.

IL

(*instruction list*) This language is a series of basic instructions. It is very close to assembly language used to program processors. Each instruction is made up of an instruction code and an operand.

implicit messaging

UDP/IP-based class 1 connected messaging for EtherNet/IP. Implicit messaging maintains an open connection for the scheduled transfer of control data between a producer and consumer. Because an open connection is maintained, each message contains primarily data, without the overhead of object information, plus a connection identifier.

INT

(*INTeger*) (encoded in 16 bits) The upper/lower limits are as follows: -(2 to the power of 15) to (2 to the power of 15) - 1.

Example: `-32768`, `32767`, `2#1111110001001001`, `16#9FA4`.

inter-controller network

An Ethernet-based network that is part of the control network, and provides data exchange between controllers and engineering tools (programming, asset management system (AMS)).

IODDT

(*input/output derived data type*) A structured data type representing a module, or a channel of a CPU. Each application expert module possesses its own IODDTs.

IP address

The 32-bit identifier, consisting of both a network address and a host address assigned to a device connected to a TCP/IP network.

IPsec

(*internet protocol security*) An open set of protocol standards that make IP communication sessions private and secure for traffic between modules using IPsec, developed by the internet engineering task force (IETF). The IPsec authentication and encryption algorithms require user-defined cryptographic keys that process each communications packet in an IPsec session.

isolated DIO network

An Ethernet-based network containing distributed equipment that does not participate in an RIO network.

L

LD

(*ladder diagram*) A programming language that represents instructions to be executed as graphical diagrams very similar to electrical diagrams (contacts, coils, etc.).

literal value of an integer

A literal value of an integer is used to enter integer values in the decimal system. Values may be preceded by the "+" and "-" signs. Underscore signs (_) separating numbers are not significant.

Example:

`-12`, `0`, `123_456`, `+986`

local rack

An M580 rack containing the CPU and a power supply. A local rack consists of one or two racks: the main rack and the extended rack, which belongs to the same family as the main rack. The extended rack is optional.

local slave

The functionality offered by Schneider Electric EtherNet/IP communication modules that allows a scanner to take the role of an adapter. The local slave enables the module to publish data via implicit messaging connections. Local slave is typically used in peer-to-peer exchanges between PACs.

M

M580 Ethernet I/O device

An Ethernet device that provides automatic network recovery and deterministic RIO performance. The time it takes to resolve an RIO logic scan can be calculated, and the system can recover quickly from a communication disruption. M580 Ethernet I/O devices include:

- local rack (including a CPU with Ethernet I/O scanner service)
- RIO drop (including an Ethernet X80 EIO adapter module)
- DRS switch with a predefined configuraton

main ring

The main ring of an Ethernet RIO network. The ring contains RIO modules and a local rack (containing a CPU with Ethernet I/O scanner service) and a power supply module.

MAST

A master (MAST) task is a deterministic processor task that is run through its programming software. The MAST task schedules the RIO module logic to be solved in every I/O scan. The MAST task has two sections:

- IN: Inputs are copied to the IN section before execution of the MAST task.
- OUT: Outputs are copied to the OUT section after execution of the MAST task.

MB/TCP

(*Modbus over TCP protocol*) This is a Modbus variant used for communications over TCP/IP networks.

MIB

(*management information base*) A virtual database used for managing the objects in a communications network. See SNMP.

Modbus

Modbus is an application layer messaging protocol. Modbus provides client and server communications between devices connected on different types of buses or networks. Modbus offers many services specified by function codes.

multicast

A special form of broadcast where copies of the packet are delivered to only a specified subset of network destinations. Implicit messaging typically uses multicast format for communications in an EtherNet/IP network.

N

network

There are two meanings:

- In a ladder diagram:
  A network is a set of interconnected graphic elements. The scope of a network is local, concerning the organizational unit (section) of the program containing the network.
- With expert communication modules:
  A network is a set of stations that intercommunicate. The term *network* is also used to define a group interconnected graphic elements. This group then makes up part of a program that may comprise a group of networks.

NIM

(*network interface module*) A NIM resides in the first position on an STB island (leftmost on the physical setup). The NIM provides the interface between the I/O modules and the fieldbus master. It is the only module on the island that is fieldbus-dependent — a different NIM is available for each fieldbus.

NTP

(*network time protocol*) Protocol for synchronizing computer system clocks. The protocol uses a jitter buffer to resist the effects of variable latency.

O

O->T

(*originator to target*) See originator and target.

ODVA

(*Open DeviceNet Vendors Association*) The ODVA supports network technologies that are based on CIP.

operation network

An Ethernet-based network containing operator tools (SCADA, client PC, printers, batch tools, EMS, etc.). Controllers are connected directly or through routing of the inter-controller network. This network is part of the control network.

originator

In EtherNet/IP, a device is considered the originator when it initiates a CIP connection for implicit or explicit messaging communications or when it initiates a message request for un-connected explicit messaging.

P

PAC

programmable automation controller. The PAC is the brain of an industrial manufacturing process. It automates a process as opposed to relay control systems. PACs are computers suited to survive the harsh conditions of an industrial environment.

port 502

Port 502 of the TCP/IP stack is the well-known port that is reserved for Modbus TCP communications.

port mirroring

In this mode, data traffic that is related to the source port on a network switch is copied to another destination port. This allows a connected management tool to monitor and analyze the traffic.

Q

QoS

(*quality of service*) The practice of assigning different priorities to traffic types for the purpose of regulating data flow on the network. In an industrial network, QoS is used to provide a predictable level of network performance.

R

rack optimized connection

Data from multiple I/O modules are consolidated in a single data packet to be presented to the scanner in an implicit message in an EtherNet/IP network.

ready device

Ethernet ready device that provides additional services to the EtherNet/IP or Modbus module, such as: single parameter entry, bus editor declaration, system transfer, deterministic scanning capacity, alert message for modifications, and shared user rights between Unity Pro and the device DTM.

RIO drop

One of the three types of RIO modules in an Ethernet RIO network. A RIO drop is an M580 rack of I/O modules that are connected to an Ethernet RIO network and managed by an Ethernet RIO adapter module. A drop can be a single rack or a main rack with an extended rack.

RIO network

An Ethernet-based network that contains 3 types of RIO devices: a local rack, an RIO drop, and a ConneXium extended dual-ring switch (DRS). Distributed equipment may also participate in an RIO network via connection to DRSs or BMENOS0300 network option switch modules.

RPI

(requested packet interval) The time period between cyclic data transmissions requested by the scanner. EtherNet/IP devices publish data at the rate specified by the RPI assigned to them by the scanner, and they receive message requests from the scanner at each RPI.

RSTP

(*rapid spanning tree protocol*) Allows a network design to include spare (redundant) links to provide automatic backup paths if an active link stops working, without the need for loops or manual enabling/disabling of backup links.

S

SCADA

(*supervisory control and data acquisition*) SCADA systems are computer systems that control and monitor industrial, infrastructure, or facility-based processes (examples: transmitting electricity, transporting gas and oil in pipelines, and distributing water).

scanner

A scanner acts as the originator of I/O connection requests for implicit messaging in EtherNet/IP, and message requests for Modbus TCP.

scanner class device

A scanner class device is defined by the ODVA as an EtherNet/IP node capable of originating exchanges of I/O with other nodes in the network.

service port

A dedicated Ethernet port on the M580 RIO modules. The port may support these major functions (depending on the module type):

- port mirroring: for diagnostic use
- access: for connecting HMI/Unity Pro/ConneXview to the CPU
- extended: to extend the device network to another subnet
- disabled: disables the port, no traffic is forwarded in this mode

SFC

(*sequential function chart*) Used to graphically represent in a structured manner the operation of a sequential CPU. This graphical description of the CPU's sequential behavior and of the various resulting situations is created using simple graphic symbols.

SFP

(*small form-factor pluggable*). The SFP transceiver acts as an interface between a module and fiber optic cables.

simple daisy chain loop

Often referred to as SDCL, a simple daisy chain loop contains RIO modules only (no distributed equipment). This topology consists of a local rack (containing a CPU with Ethernet I/O scanner service), and one or more RIO drops (each drop containing an RIO adapter module).

SMTP

(*simple mail transfer protocol*) An email notification service that allows controller-based projects to report alarms or events. The controller monitors the system and can automatically create an email message alert with data, alarms, and/or events. Mail recipients can be either local or remote.

SNMP

(*simple network management protocol*) Protocol used in network management systems to monitor network-attached devices. The protocol is part of the internet protocol suite (IP) as defined by the internet engineering task force (IETF), which consists of network management guidelines, including an application layer protocol, a database schema, and a set of data objects.

SNTP

(*simple network time protocol*) See NTP.

SOE

(*sequence of events*) The process of determining the order of events in an industrial system and correlating those events to a real-time clock.

ST

(*structured text*) The structured literal language is a developed language similar to computer programming languages. It can be used to organize a series of instructions.

sub-ring

An Ethernet-based network with a loop attached to the main ring, via a dual-ring switch (DRS) or BMENOS0300 network option switch module on the main ring. This network contains RIO or distributed equipment.

subnet mask

The 32-bit value used to hide (or mask) the network portion of the IP address and thereby reveal the host address of a device on a network using the IP protocol.

switch

A multi-port device used to segment the network and limit the likelihood of collisions. Packets are filtered or forwarded based upon their source and destination addresses. Switches are capable of full-duplex operation and provide full network bandwidth to each port. A switch can have different input/output speeds (for example, 10, 100 or 1000Mbps). Switches are considered OSI layer 2 (data link layer) devices.

T

T->O

(*target to originator*) See target and originator.

target

In EtherNet/IP, a device is considered the target when it is the recipient of a connection request for implicit or explicit messaging communications, or when it is the recipient of a message request for un-connected explicit messaging.

TCP

(*transmission control protocol*) A key protocol of the internet protocol suite that supports connection-oriented communications, by establishing the connection necessary to transmit an ordered sequence of data over the same communication path.

TCP/IP

Also known as *internet protocol suite*, TCP/IP is a collection of protocols used to conduct transactions on a network. The suite takes its name from two commonly used protocols: transmission control protocol and internet protocol. TCP/IP is a connection-oriented protocol that is used by Modbus TCP and EtherNet/IP for explicit messaging.

TFTP

(*trivial file transfer protocol*) A simplified version of *file transfer protocol* (FTP), TFTP uses a client-server architecture to make connections between two devices. From a TFTP client, individual files can be uploaded to or downloaded from the server, using the user datagram protocol (UDP) for transporting data.

TIME_OF_DAY

See `TOD`.

TOD

(*time of day*) The `TOD` type, encoded in BCD in a 32-bit format, contains this information:

- the hour encoded in an 8-bit field
- the minutes encoded in an 8-bit field
- the seconds encoded in an 8-bit field

NOTE: The eight least significant bits are not used.

The TOD type is entered in this format: xxxxxxxx: **`TOD#`**`<Hour>:<Minutes>:<Seconds>`

This table shows the upper/lower limits of each field:

Field	Limits	Comment
Hour	[00,23]	The leading 0 is displayed; it can be omitted during data entry.
Minute	[00,59]	The leading 0 is displayed; it can be omitted during data entry.
Second	[00,59]	The leading 0 is displayed; it can be omitted during data entry.

Example: `TOD#23:59:45`.

TR

(*transparent ready*) Web-enabled power distribution equipment, including medium- and low-voltage switch gear, switchboards, panel boards, motor control centers, and unit substations. Transparent Ready equipment allows you to access metering and equipment status from any PC on the network, using a standard web browser.

trap

A trap is an event directed by an SNMP agent that indicates one of these events:

- A change has occurred in the status of an agent.
- An unauthorized SNMP manager device has attempted to get data from (or change data on) an SNMP agent.

U

UDP

(*user datagram protocol*) A transport layer protocol that supports connectionless communications. Applications running on networked nodes can use UDP to send datagrams to one another. Unlike TCP, UDP does not include preliminary communication to establish data paths or provide data ordering and checking. However, by avoiding the overhead required to provide these features, UDP is faster than TCP. UDP may be the preferred protocol for time-sensitive applications, where dropped datagrams are preferable to delayed datagrams. UDP is the primary transport for implicit messaging in EtherNet/IP.

UMAS

(*Unified Messaging Application Services*) UMAS is a proprietary system protocol that manages communications between Unity Pro and a controller.

UTC

(*coordinated universal time*) Primary time standard used to regulate clocks and time worldwide (close to former GMT time standard).

V

variable

Memory entity of type `BOOL`, `WORD`, `DWORD`, etc., whose contents can be modified by the program currently running.

VLAN

(*virtual local area network*) A local area network (LAN) that extends beyond a single LAN to a group of LAN segments. A VLAN is a logical entity that is created and configured uniquely using applicable software.

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